

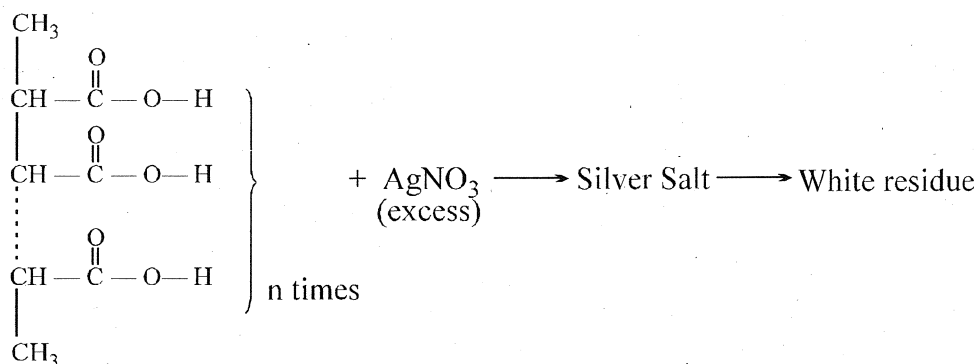
PHYSICAL CHEMISTRY

TIME : 30 MIN.

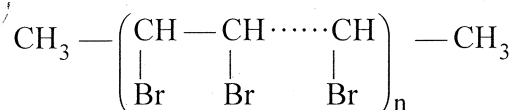
SECTION-I : (i) Only One option correct Type

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(-1)**

- 20 mL of 0.1 M solution of compound $\text{Na}_2\text{CO}_3 \cdot \text{NaHCO}_3 \cdot 2\text{H}_2\text{O}$ is titrated against 0.05 M HCl, x mL of HCl is used when phenolphthalein is used as an indicator and y mL of HCl is used when methyl orange is the indicator in two separate titrations. Hence (y - x) is :-
(A) 40 mL (B) 80 mL (C) 120 mL (D) none of these
- A 150 mL of solution of I_2 is divided into two unequal parts. I part reacts with hypo solution in acidic medium. 15 mL of 0.4 M hypo was consumed. II part was added with 100 mL of 0.3 M NaOH solution. Residual base required 10 mL of 0.3 M H_2SO_4 solution for complete neutralization. What was the initial concentration of I_2 ?
(A) 0.08 M (B) 0.1 M (C) 0.2 M (D) none of these
- 25 mL of household bleach solution was mixed with 30 mL of 0.50 M KI and 10 mL of 4 N acetic acid. In the titration of the liberated iodine, 48 mL of 0.25 N $\text{Na}_2\text{S}_2\text{O}_3$ was used to reach the end point. The molarity of the household bleach solution is
(A) 0.48 M (B) 0.96 M (C) 0.24 M (D) 0.024 M
- A volume of 100 ml of 0.1 M $\text{NaAl(OH)}_2\text{CO}_3$ is neutralized by 0.25 N - HCl to form NaCl, AlCl_3 and CO_2 . Volume of HCl required is
(A) 10 ml (B) 40 ml (C) 100 ml (D) 160 ml
- Chloride samples are prepared for analysis by using NaCl, KCl and NH_4Cl separately or as mixture. What minimum volume of 5 % by weight AgNO_3 solution (sp.gr. = 1.7) must be added to a sample of 1.07 g in order to ensure complete precipitation of chloride in every possible case?
(A) 36.58 ml (B) 40.00 ml (C) 28.72 ml (D) 105.3 ml
- Consider the following set of reactions



If 0.1 moles of silver salt is taken & wt. of residue obtained is 54 gms then what will be the molecular mass of



Given : (Ag = 108, Br = 80)

- (A) 495 (B) 402 (C) 465 (D) 525

7. A mixture of FeO and Fe_2O_3 is reacted with acidified KMnO_4 solution having a concentration of $2/5 \text{ M}$, 100 mL of which was used. The solution was then titrated with Zn dust which converted Fe^{3+} of the solution to Fe^{2+} . The Fe^{2+} required 1000 mL of $2/15 \text{ M}$ $\text{K}_2\text{Cr}_2\text{O}_7$ solution. The % by mol of FeO in the original mixture is -
 (A) 0.2 % ~~(B) 40%~~ (C) 60% ~~(D) 25%~~
8. H_2O_2 is reduced rapidly by Sn^{2+} , the products being Sn^{4+} and water. H_2O_2 decomposes slowly at room temperature to yield O_2 and water. Calculate the volume of O_2 produced at 273 K and 1.00 atm when 200 g of 10.0 % by mass H_2O_2 in water is treated with 88.2 mL of 1 M Sn^{2+} and then the mixture is allowed to stand until no further reaction occurs.
 (A) 2.8 L ~~(B) 5.6 L~~ (C) 11.2 L (D) 22.4 L
9. 80 gm of a sample of Anhydrous CuSO_4 was dissolved in water and made to 250 mL. 25 mL of this solution after taking usual precaution was treated with a little excess of KI solution. A white ppt. of Cu_2I_2 and iodine was evolved. The iodine so evolved required 40 mL of 1 M hypo solution. What is the percentage purity of CuSO_4 solution. [Mol. wt. of $\text{CuSO}_4 = 160$]
 (A) 64 % (B) 40% ~~(C) 80%~~ (D) 50%
10. A 458 g sample containing Mn_3O_4 was dissolved and all manganese was converted to Mn^{2+} . In the presence of fluoride ion, Mn^{2+} is titrated with 3 lit of KMnO_4 solution (which was 1.25 N against oxalate in acidic medium), both reactants being converted to a complex of Mn(III), What was the % of Mn_3O_4 in the sample?
~~(A) 50 %~~ (B) 75% (C) 25% (D) 40%
11. A 2.024 g sample containing $\text{Ba}(\text{SCN})_2$ was dissolved in a bicarbonate solution. 50.0 mL of 1 N iodine solution was added, and the mixture was allowed to stand for five minutes. Iodine oxidises SCN^- into SO_4^{2-} and HCN . The solution was then acidified, and the excess I_2 was titrated with 26 mL of 1 M sodium thiosulphate. The percent $\text{Ba}(\text{SCN})_2$ in sample.
~~(A) 25%~~ (B) 50% (C) 12.5% (D) 20%
12. A 3.00 g sample containing Fe_3O_4 , Fe_2O_3 and an inert impure substance, is treated with excess of KI solution in presence of dilute H_2SO_4 . The entire iron is converted into Fe^{2+} along with the liberation of iodine. The resulting solution is diluted to 100 mL. A 20 mL of the diluted solution require 11 mL of 0.5 M $\text{Na}_2\text{S}_2\text{O}_3$ solution to reduce the iodine present. A 50 mL of diluted solution after complete extraction of the iodine requires 12.80 mL of 0.25 KMnO_4 solution in dilute H_2SO_4 medium for the oxidation of Fe^{2+} . Calculate the percentages of inert impurity in the original sample is.
 (A) 49.33% (B) 34.8% ~~(C) 15.87%~~ (D) 8.73%
13. Hydrogen peroxide solution (20 mL) reacts quantitatively with a solution of KMnO_4 (20 mL) acidified with dilute H_2SO_4 . The same volume of KMnO_4 solution is just decolorized by 10 mL of MnSO_4 in neutral medium simultaneously forming a dark brown precipitate of hydrated MnO_2 . The brown precipitate is dissolved in 10 mL of 0.2 M sodium oxalate under boiling condition in the presence of dilute H_2SO_4 . The molarity of H_2O_2 solution is -
~~(A) 0.1M~~ (B) 0.05M (C) 0.2M% (D) 0.5M
14. For the estimation of nitrogen, 1.4 g of an organic compound was digested by Kjeldahl method and the evolved ammonia was absorbed in 60 mL of $\frac{\text{M}}{10}$ sulphuric acid. The unreacted acid required 20 mL of $\frac{\text{M}}{10}$ sodium hydroxide for complete neutralization. The percentage of nitrogen in the compound is :
 (A) 3% (B) 5% (C) 6% ~~(D) 10%~~

15. X gm A atoms on combining with Y atoms of B form 5 molecules of a compound containing A & B. Find the molecular weight of compound formed. (Atomic weight of B = M)
- (A) $\frac{(XN_A + MY)}{5}$ (B) $\frac{X + M}{5}$ (C) $\frac{X + MY}{5}$ (D) $\left(\frac{X + MYN_A}{5}\right)$
16. A mixture of CH_4 & air (containing 80% N_2 20% O_2 by volume) of a composition such that when underwent combustion gave maximum heat (assume combustion of only CH_4). Then which of the statements are correct, regarding composition of initial mixture. (X presents mole fraction)
- (A) $X_{\text{CH}_4} = \frac{1}{11}, X_{\text{O}_2} = \frac{2}{11}, X_{\text{N}_2} = \frac{8}{11}$ (B) $X_{\text{CH}_4} = \frac{3}{8}, X_{\text{O}_2} = \frac{1}{8}, X_{\text{N}_2} = \frac{1}{2}$
- (C) $X_{\text{CH}_4} = \frac{1}{6}, X_{\text{O}_2} = \frac{1}{6}, X_{\text{N}_2} = \frac{2}{3}$ (D) Data insufficient
17. Similar to the % labelling of oleum, a mixture of H_3PO_4 and P_4O_{10} is labelled as $(100 + x)\%$ where x is the maximum mass of water which can react with P_4O_{10} present in 100 gm mixture of H_3PO_4 and P_4O_{10} . If such a mixture is labelled as 127 %. Mass of P_4O_{10} in 100 gm of mixture, is
- (A) 71 gm (B) 47 gm (C) 83 gm (D) 35 gm
18. For the estimation of nitrogen, 1.4 g of an organic compound was digested by Kjeldahl method and the evolved ammonia was absorbed in 60 mL of $\frac{M}{10}$ sulphuric acid. The unreacted acid required 20 mL of $\frac{M}{10}$ sodium hydroxide for complete neutralization. The percentage of nitrogen in the compound is :
- (A) 3% (B) 5% (C) 6% (D) 10%
19. 200 ml of a gaseous mixture containing CO , CO_2 and N_2 on complete combustion in just sufficient amount of O_2 showed contraction of 40 ml. When the resulting gases were passed through KOH solution it reduces by 50 % then the volume ratio of $V_{\text{CO}_2} : V_{\text{CO}} : V_{\text{N}_2}$ in original mixture is.
- (A) 4 : 1 : 5 (B) 2 : 3 : 5 (C) 1 : 4 : 5 (D) 1 : 3 : 5
20. Two isotopes of an element Q are Q^{97} (23.4% abundance) and Q^{94} (76.6% abundance). Q^{97} is 8.082 times heavier than C^{12} and Q^{94} is 7.833 times heavier than C^{12} . What is the average atomic weight of the element Q?
- (A) 94.702 (B) 78.913 (C) 96.298 (D) 94.695

(ii) One or more options correct Type

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. 4(-1)

21. When $\text{N}_2(\text{g})$ and $\text{H}_2(\text{g})$ are mixed $\text{N}_2\text{H}_4(\text{g})$, $\text{NH}_3(\text{g})$ or both may form, depending upon the relative amount of N_2 and H_2 taken. If initial moles of N_2 , H_2 are x, y and final moles of N_2H_4 , NH_3 are z, v, then the correct options from the following in order of (x, y, z, v) is/are -
- (A) (2, 2, 1, 0) (B) (3, 8, 1, 4) (C) (4, 9, 4, 1) (D) (0.5, 3, 0, 1)
22. A definite volume of pure ammonia (NH_3) gas is passed through a series of electric sparks by which the volume becomes 90 ml. The increase in volume is due to formation of nitrogen (N_2) and hydrogen (H_2) gases. All the gases finally present are washed with dilute sulphuric acid solution, by which the volume of gases becomes 80 ml. All the volume are measured at the same temperature and pressure. Which of the following statement(s) is/are correct regarding the original ammonia sample?
- (A) The volume of NH_3 gas taken was 40 ml
- (B) The volume of NH_3 gas taken was 50 ml
- (C) Only 80% of NH_3 gas decomposed into N_2 and H_2 gases.
- (D) Only 20% of NH_3 gas decomposed into N_2 and H_2 gases.

23. A quantity of 5.68 g of pure P_4O_{10} is dissolved completely in sufficient water and the solution is diluted to 250 ml. Which of the following statement(s) is/are correct?
- (A) The diluted solution has molarity 0.32 with respect to H_3PO_4 .
(B) 25 ml of the diluted solution exactly requires 48 ml of 0.5 M – NaOH solution for complete neutralization.
(C) 15 ml of the diluted solution exactly requires 36 ml of 0.2 M – $BaCl_2$ solution for complete precipitation of phosphate.
(D) 40 ml of the diluted solution exactly requires 48 ml of 0.8 N – KOH solution for the first equivalent point.
24. 24 mL of a solution containing HCl was treated with excess of 0.004 M KIO_3 and KI solution of unknown concentration where I_2 liberated is titrated against a standard solution of 0.02 M $Na_2S_2O_3$ solution whose 24 mL were used up. Identify the correct informations ?
- (A) volume of KIO_3 solution consumed = 20 ml
(B) Molarity of HCl solution was 0.2 M
(C) Normality of HCl solution for this reaction = $\frac{0.1}{6}$ N
(D) Moles of I_2 formed in first reaction = 0.24
25. The equivalent volume of a gaseous substance is 5.6 L at $0^\circ C$ and 1 atm. The substance may be
- (A) CH_4 gas in combustion (B) O_3 gas as oxidizing agent
(C) H_2S gas as reducing agent (D) CO_2 formed from carbon

SECTION-IV : (Integer Value Correct Type)

This section contains **5 questions**. The answer to each question is a **single digit Integer**, ranging from **0 to 9** (both inclusive) **4(-1)**

1. A sample of crude copper is boiled in H_2SO_4 till all the copper has reacted. The impurities are inert to the acid. The SO_2 liberated in the action is passed into 100 mL of 0.4 M acidified $KMnO_4$ ($SO_2 \rightarrow SO_4^{2-}$). The solution of $KMnO_4$ after passage of SO_2 is allowed to react with oxalic acid and requires 25 mL of 1 M oxalic acid. If the purity of copper is 95.25%, what was the weight of crude sample (in gm). (5)
2. Equivalent weight of Br_2 is 96 in the following disproportionation reaction:
 $Br_2 + OH^- \rightarrow Br^- + H_2O + ?$ (oxidized product) (5)
The oxidation state of Br in the oxidized product is (Br = 80)
3. Basic solution of Na_2XeO_6 is powerful oxidants. How many millimoles of $Mn(NO_3)_2 \cdot 6H_2O$ reacts with 62.5 ml of a 0.04 M basic solution of Na_4XeO_6 that contains an excess of sodium hydroxide if the products include Xe and a solution of sodium permanganate? (Mn = 55) (4)
4. A mixture of Xe and F_2 was heated. A sample of white solid thus formed reacted with hydrogen to give 56 ml of Xe at $0^\circ C$ and 1 atm and HF formed required 60 ml of 0.25 M – NaOH for complete neutralization. If the molecular formula of the solid formed is XeF_x , then the value of x is (6)
5. When ammonium vanadate is heated with oxalic acid solution, a substance Z is formed. A sample of Z was treated with $KMnO_4$ solution in hot acidic solution. The resulting liquid was reduced with SO_2 , the excess SO_2 boiled off and the liquid again titrated with same $KMnO_4$. The ratio of the volumes of $KMnO_4$ used in the two titrations was 5 : 1. What is the oxidation state of vanadium in substance Z? Given that $KMnO_4$ oxidize all oxidation state of vanadium to vanadium (+5) and SO_2 reduces V(+5) to V(+4) (2)

SECTION-I : (ii) One or more options correct Type

This section contains **9 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. 4(-1)

- Total number of protons, neutrons and electrons present in 14 mg of ${}^6\text{C}^{14}$ is
(Take $N_A = 6 \times 10^{23}$)
(A) 1.2×10^{22} (B) 1.2×10^{25} (C) 7.2×10^{21} (D) 1.08×10^{22}
- Which of the following has the smallest no. of molecules :-
(A) 22.4 mL of CO_2 gas at 1 atm 273K $\frac{10^{-3}}{2} N_A$ (B) 22 g of CO_2 gas $\frac{1}{2} N_A$
(C) 11.2 L of CO_2 gas at 1 atm 273K $\frac{1}{2} N_A$ (D) 0.1 moles of CO_2 gas $\frac{1}{10} N_A$
- A compound contains 69.5% oxygen and 30.5% nitrogen and its molecular weight is 92. The formula of that compound is :-
(A) N_2O (B) NO_2 (C) N_2O_4 (D) N_2O_5
- 26.8 g of $\text{Na}_2\text{SO}_4 \cdot n\text{H}_2\text{O}$ contains 12.6 g of water. The value of 'n' is :-
(A) 1 (B) 10 (C) 6 (D) 7
- Which of the following series of compounds have same mass percentage of carbon ?
(A) CO_2 , CO (B) CH_4 , C_2H_6 , C_2H_2
(C) C_2H_2 , C_6H_6 , C_{10}H_8 (D) HCHO , CH_3COOH , $\text{C}_6\text{H}_{12}\text{O}_6$
- If a mole is defined to be 3×10^{24} (instead of Avogadro's Number, 6×10^{23}), what would be the mass of one mole of Argon atoms ? [Atomic mass of Ar on conventional scale is 40]
(A) 40 gm (B) 200 gm (C) 20 g (D) 8 gm
- 400 gm mixture of AB and AC_2 contains 26.25% of "A" by mass. Select the correct statement(s).
(Atomic masses of A = 15, B = 25, C = 22.5)
(A) % mass of AB is 10 (B) Number of moles of AC_2 in the mixture is 4
(C) Number of moles of AB is 1 (D) Total moles of "A atoms" in mixture is 7
- Choose the incorrect statement(s)
(A) 1 gm-molecule always contains same number of atoms
(B) weight of one molecule in gm is equal to its molar mass $\frac{1}{N_A}$
(C) Number of atoms in 2 gm of hydrogen is greater than 11.35 litre hydrogen at STP
(D) Volume of 16 gm oxygen gas at 2 atm, 300 K is greater than volume of 2 gm hydrogen gas at 1 atm, 300K.
- In 2.6 kg of $\text{FeSO}_4 \cdot 6\text{H}_2\text{O}$ (At. wt of Fe = 56)
(A) No. of atom of O are $100 \times N_A$ (B) Moles of H atoms are $120 \times N_A$
(C) Molecules of water are $6 \times N_A$ (D) Moles of e^- present in SO_4^{2-} are 500

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SECTION-II : Matrix-Match Type

This Section contains **1 question**. Question has **four statements** (A, B, C and D) given in **Column I** and five statements (P, Q, R, S and T) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question, against statement B, darken the bubbles corresponding to Q and R in the ORS.

8(0)

1.	Column-I (Amount of substance)	Column-II (No. of moles of particular atoms in the given substance)
(A)	6.022×10^{24} molecules of $\text{Al}_2(\text{SO}_4)_3 \cdot 3\text{H}_2\text{O}$ QTS	(P) 15 mole O-atoms
(B)	90 gm $\text{C}_6\text{H}_{12}\text{O}_6$ R	(Q) 60 mole H-atoms
(C)	112 litre SO_3 (g) at 1 atm and 0°C . P	(R) 3 mole O-atoms
(D)	3.75 mole N_2O_4 (g) P	(S) 30 mole S-atoms
		(T) 150 mole O-atoms

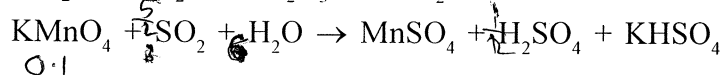
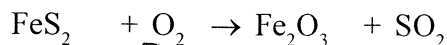
SECTION-I : (ii) One or more options correct Type

This section contains **11 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. 4(-1)

- For reaction ; $\text{MnO}_2 + 4\text{HCl} \longrightarrow \text{MnCl}_2 + 2\text{H}_2\text{O} + \text{Cl}_2(\text{g})$
261 gm MnO_2 mixed with 448 litre of HCl gas at 273°C & 1 atm pressure to produce product.
[$N_A = 6 \times 10^{23}$; Atomic mass of Mn = 55, Cl = 35.5]
Select correct statement(s).
 (A) MnO_2 is limiting reagent (B) Chlorine gas produced contains 15×10^{23} molecules.
 (C) Moles of excess reactant left is 0.5 moles (D) If % yield of reaction is 50%, then mass of MnCl_2 obtained will be 315.
- A regular cube of metal measures exactly 10 cm on an edge and has density 8g/cc. If the cube contains 6×10^{25} atoms of the metal, determine atomic weight of metal ? (take $N_A = 6 \times 10^{23}$)
 (A) 40 (B) 60 (C) 80 (D) 100
- 10 mole of A_2B_3 contains 100gm A atom & 60 gm B atoms. Choose the correct statements -
 (A) Molecular weight of A_2B_3 is equal to 16 (B) Atomic weight of A is equal to 5
 (C) Weight of one atom of B is equal to 2 (D) Atomic weight of B is equal to 6
- Aspartame an artificial sweetner contains 10 % by weight of nitrogen. There are two nitrogen atoms per molecule, then the molecular weight of aspartame is -
 (A) 560 (B) 70 (C) 140 (D) 280
- A compound of X and Y has equal mass of them. If their atomic weights are 30 and 20 respectively. Molecular formula of that compound (its mol. wt. is 120) could be -
 (A) X_2Y_2 (B) X_3Y_3 (C) X_2Y_3 (D) X_3Y_2
- Number of neutrons in 5.5 gm T_2O (T is ${}_1\text{H}^3$) are -
 (A) $0.25 N_A$ (B) $2.5 N_A$ (C) $3 N_A$ (D) $4 \times N_A$
- A compound contains 38.8% C, 16.0% H and 45.2% N. The empirical formula of the compound would be
 (A) CH_3NH_2 (B) CH_3CN (C) $\text{C}_2\text{H}_5\text{CN}$ (D) $\text{CH}_2(\text{NH})_2$
- 60 gm of an organic compound containing C, H & O atoms on complete combustion gave 88gm CO_2 & 36gm H_2O . The empirical formula of the organic compound is :
 (A) $\text{C}_2\text{H}_2\text{O}$ (B) CHO (C) CH_2O (D) CHO_2
- Volume of O_2 obtained at 2 atm & 546K, by the complete decomposition of 8.5 g NaNO_3 is
 $2\text{NaNO}_3 \rightarrow 2\text{NaNO}_2 + \text{O}_2$
 (A) 2.24 lit (B) 1.12 lit (C) 0.84 lit (D) 0.56 lit
- Consider the following reaction
 $2\text{Na} + 2\text{NH}_3 \longrightarrow 2\text{NaNH}_2 + \text{H}_2(\text{g})$
 $2\text{NaNH}_2 + \text{C} \longrightarrow \text{Na}_2\text{CN}_2 + 2\text{H}_2(\text{g})$
 $\text{Na}_2\text{CN}_2 + \text{C} \longrightarrow 2\text{NaCN}$
 51gm dry ammonia gas is passed over excess heated sodium to form sodamide (NaNH_2) which is further reacted with carbon (excess) to finally form NaCN . Find the total volume of $\text{H}_2(\text{g})$ evolved at 0.5 atm, 273K.
 (A) 201.6 L (B) 100.8 L (C) 403.2 L (D) 50.4 L

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11. Find the moles of iron pyrite (FeS_2), sufficient to produce enough SO_2 on roasting (heating in excess of O_2) such that it (SO_2) completely decolourize a 1L solution of KMnO_4 containing 15.8 g/L of it. The equation are - [$\text{Mn} = 55$, $\text{K} = 39$, $\text{S} = 32$]



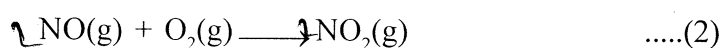
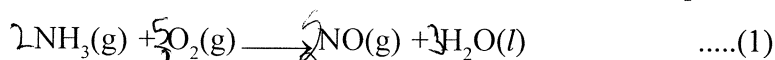
- (A) 0.1 mol (B) 0.5 mol (C) $\frac{0.5}{2}$ mol (D) $\frac{0.5}{4}$ mol

(iii) Paragraph Type

This section contains **1 paragraphs** each describing theory, experiment, data etc. **3 questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Paragraph for Q.13 to Q.15

The following sequence of reaction occurs in commercial production of aqueous nitric acid.



In an industry 378 kg of HNO_3 was required to be produced

13. If % yield all reactions is 100% then mass (kg) of NH_3 required is (NO produced in reaction-3 is not re-used in reaction-2)
(A) 306 (B) 170 (C) 126 (D) 153
14. If % yield of reaction (1), (2) & (3) is 85%, 60% & 50% respectively. Then mass (kg) of NH_3 required (NO produced in reaction-3 is not re-used in reaction-2)
(A) 30 (B) 600 (C) 6000 (D) 300
15. If % yield all reactions is 100% then mass (kg) of NH_3 required is (NO produced in reaction-3 is re-used in reaction-2)
(A) 153 (B) 126 (C) 102 (D) can't be determined.

SECTION-IV : (Integer Value Correct Type)

This section contains **2 questions**. The answer to each question is a **single digit Integer**, ranging from **0 to 9** (both inclusive) **4(-1)**

1. 2.0 g of a sample containing NaCl , NaBr and some inert impurity is dissolved in enough water and treated with excess of AgNO_3 solution. A 3.0g of precipitate was formed. Precipitate on shaking with aqueous NaBr gain 0.76 g of weight. Determine mass percentage of NaCl in the original sample. *24.8%*

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

2. In an organic compound of molar mass 108 g mol^{-1} C, H and N atoms are present in 9 : 1 : 3.5 by weight. If molecular formula is represented by $\text{C}_x\text{H}_y\text{N}_z$, then ratio of x : y is *3*

(i) Only One option correct Type

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(-1)**

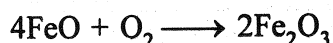
1. If a solution of stearic acid ($C_{18}H_{36}O_2$; $M = 284$) in benzene contains 1.42 gm acid per L. When this solution (100 L) is dropped on surface, C_6H_6 gets evaporated and acid forms a unimolecular layer on the surface. If it covers an area 6020 cm^2 with unimolecular film. Find the area covered by one molecule of acid.

☒ (A) $2 \times 10^{-20} \text{ cm}^2$ (B) $4 \times 10^{-20} \text{ cm}^2$ (C) $2 \times 10^{20} \text{ cm}^2$ (D) $4 \times 10^{20} \text{ cm}^2$

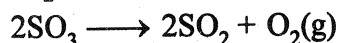
2. A compound containing 5 gm sulphur and 5gm oxygen atom. Then empirical formula of compound is :

(A) SO ☒ (B) SO_2 (C) S_2O (D) SO_3

3. Ferric oxide can be obtained by oxidation of FeO :



The O_2 gas required can be prepared by the following reaction.



What is the maximum amount of Fe_2O_3 can be produced by 144 g FeO and 160 g SO_3 .

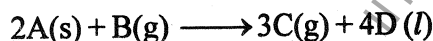
[Atomic mass of Fe = 56]

(A) 0.320 g (B) 80 g (C) 120 g ☒ (D) 160 g

4. A mixture containing 3 moles each of C_4H_8 and C_6H_6 undergoes complete combustion with O_2 to form CO_2 and H_2O . Calculate total mass of CO_2 produced

☒ (A) 1320 gm (B) 610 gm (C) 528 gm (D) 792 gm

5. Select the correct statements (s) for the following reaction



(A) 2 moles of A always produce 3 moles of C & 4 moles of D

☒ (B) 22.4 l of B (g) produces 3 moles of C at 1 atm & 273K in excess of A.

(C) B will always remain in excess because volume of gas is very high than solid

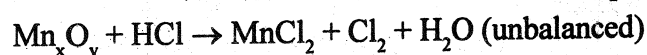
(D) Moles of D produced is always less than C because volume of liquid is less than gas.

(iii) Paragraph Type

This section contains **1 paragraphs** each describing theory, experiment, data etc. **2 questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Paragraph for Question 07 & 08

An unknown compound "A" (Mn_xO_y) composed of manganese and oxygen, has 36.7 % oxygen by weight. When 8.7 of "A" is heated with HCl it liberates Cl_2 gas as per the following reaction :



7. The simplest formula of A is :

(A) MnO ☒ (B) MnO_2 (C) Mn_2O_3 (D) Mn_3O_4

8. The volume of Cl_2 gas at STP obtained when 8.7 gm of compound A is heated with excess of HCl (assume molecular formula & empirical formula to be same) -

(A) 1.12 L ☒ (B) 2.24 L (C) 4.48 L (D) 0.56 L

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(iv) Matching List Type

This Section contains **1 multiple choice questions**. Each question has **matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **4(0)**

9. Column-I (Amount of substance)	Column-II (No. of moles of particular atoms in the given substance)
(P) 6.022×10^{24} molecules of $\text{Al}_2(\text{SO}_4)_3 \cdot 3\text{H}_2\text{O}$	(1) 15 mole O-atoms
(Q) 90 gm $\text{C}_6\text{H}_{12}\text{O}_6$	(2) 3 mole O-atoms
(R) 112 litre $\text{SO}_3(\text{g})$ at 1 atm and 0°C .	(3) 2.5 mole O-atoms
(S) 54 gram $\text{N}_2\text{O}_5(\text{g})$	(4) 150 mole O-atoms

Code :

	(P)	(Q)	(R)	(S)
(A)	4	2	1	3
(B)	4	2	3	1
(C)	4	1	2	3
(D)	2	1	4	3

SECTION-II : Matrix-Match Type

This Section contains **1 question**. Question has **four statements** (A, B, C and D) given in **Column I** and five statements (P, Q, R, S and T) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question, against statement B, darken the bubbles corresponding to Q and R in the ORS. **8(0)**

1. Match the column :

Column - I	Column - II
(A) N_2 <i>R, S</i>	(P) 40% carbon by mass
(B) CO <i>R, S</i>	(Q) Empirical formula CH_2O
(C) $\text{C}_6\text{H}_{12}\text{O}_6$ <i>T, Q, P</i>	(R) Vapour density = 14
(D) CH_3COOH <i>Q, T, P</i>	(S) $14N_A$ electrons in a mole (N_A = Avogadro no.)
	(T) 25 % carbon by mole

SECTION-IV : (Integer Value Correct Type)

This section contains **2 questions**. The answer to each question is a **single digit Integer**, ranging from **0 to 9** (both inclusive) **4(-1)**

- Calculate moles of electron in 1900 mg of PO_4^{3-} ion. *1*
- A 96 gm mixture containing CaCO_3 and MgCO_3 on heating produces a gas which dissolves in 1 litre of water to form 1 M H_2CO_3 solution. Mole ratio of CaCO_3 & MgCO_3 in original mixture. *3*

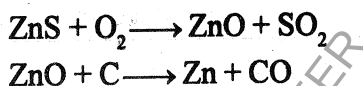
SECTION-I : (i) Only One option correct Type

This section contains 4 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. 3(-1)

- 300 gm, 30% (w/w) NaOH solution is mixed with 500 gm 40% (w/w) NaOH solution. What is % (w/v) NaOH if density of final solution is 2 gm/mL :
(A) 72.5 (B) 65 (C) 62.5 (D) 36.25
- A container contains O₂, He & SO₂ gases such that mass of SO₂ is double the mass of O₂ & mass of He is half the mass of O₂ gas. The mole fraction of SO₂ gas in the mixture is

(A) $\frac{1}{6}$ (B) $\frac{2}{3}$ (C) $\frac{5}{6}$ (D) $\frac{3}{2}$

- If mole fraction of NaCl_(aq) is same as that of water then molality of NaCl is
(A) 5.55 (B) 55.55 (C) 0.18 (D) 58.5
- The following sequence of reaction may be used to extract zinc from its sulphide ore :



How many tons of Zn (approx.) can be obtained from 10 tons of ZnS assuming that the yield is 75% ?

[Atomic weight of Zn = 65, S = 32]

(A) 5 (B) 10 (C) 12 (D) 15

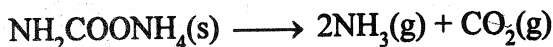
(iii) Paragraph Type

This section contains 1 paragraphs each describing theory, experiment, data etc. 3 questions relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has only one correct answer among the four choices (A), (B), (C) and (D). 3(0)

Paragraph for Q.6 to Q.8

81 gm mixture of MgCO₃(s) and NH₂COONH₄(s) is heated to constant mass. If vapour density of gaseous mixture evolved was found to be 61/4 then

Given that :



- Mole % of MgCO₃ in original sample -
(A) 50% (B) 60% (C) 75% (D) 33.33 %
- Volume (in litre) of the total gases produced at 1 atm & 273K -
(A) 22.4 (B) 44.8 (C) 179.2 (D) 89.6
- If above gaseous mixture at 273K is passed through KOH solution, contraction in volume (in litre) will be (Assume KOH solution absorb CO₂ gas completely)-
(A) 11.2 (B) 22.4 (C) 44.8 (D) 89.6

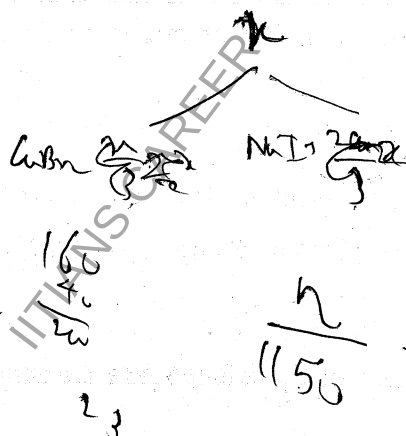
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SECTION-IV : (Integer Value Correct Type)

This section contains 4 questions. The answer to each question is a single digit Integer, ranging from 0 to 9 (both inclusive) 4(-1)

- 6 gm nitrogen on successive reaction with different compounds gets finally converted into 30 gm $[\text{Cr}(\text{NH}_3)_x\text{Br}_2]$ value of x is - $\times$
[Atomic mass of Cr = 52, Br = 80]
- Calculate the density (in gm/ml) of aqueous NaOH solution of which molarity & (w/w%) are equal 4
- If 40 gm of NaOH is added in 2 litre solution of 4M NaOH. Find the molality of final solutions if its density is 0.68gm/ml. 9
- A mixture of NaI and CaBr_2 having mole fraction of CaBr_2 equal to $\frac{1}{3}$. To this mixture water is added and solution is made upto 1 litre where molality of NaI is found to be $\frac{10}{9}$. If density of solution is 1.15 g/ml. Calculate mole of NaI in original mixture. |
[Atomic mass of I = 127, Br = 80]

$2 \text{ litre} \times 0.68$
1360

$571 = 9$
 $\frac{360}{9}$
 $\frac{160}{2}$



$1150 - \frac{200n}{3} - \frac{150 \times 2n}{3} = \frac{10}{9}$

$3450 - 600n - 300n \times \frac{5}{9} = \frac{2n}{3}$

$3450 - 500n = 9n$

$12250 = 2500n$

PHYSICAL CHEMISTRY

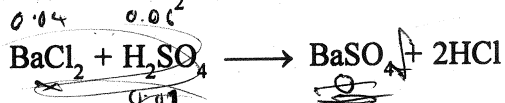
TIME : 45 MIN.

SECTION-I : (ii) One or more options correct Type

This section contains 2 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which ONE or MORE are correct. 4(-1)

1. 100 ml aqueous solution containing equimolar mixture of Ca(OH)_2 and Al(OH)_3 requires 0.5 litre of 4M HCl for complete neutralisation. Molarity of Ca(OH)_2 in the original solution is :
(A) 2 M (B) 4 M (C) 8 M (D) 10 M

2. 400 ml 0.1 M BaCl_2 is mixed with 600 ml 0.1 M H_2SO_4 to form product according to following reaction



Select the correct option(s), after reaction is completed

- precipitate*
(A) Molarity of Ba^{+2} ions in final solution is 0.04 M
(B) Molarity of SO_4^{2-} ions in final solution is 0.02 M
(C) Molarity of H^+ ions do not change on mixing
(D) Final molarity of H^+ ion in solution is 0.12 M

(iii) Paragraph Type

This section contains 1 paragraphs each describing theory, experiment, data etc. 3 questions relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has only one correct answer among the four choices (A), (B), (C) and (D). 3(0)

Paragraph for Questions 3 to 5

Air sample from an industrial town, heavily polluted by CO_2 was collected and analyzed. In one analysis, 56 L of air measured at 1atm & 273K was passed through a 250 mL of 0.025M NaOH solution, where $\text{CO}_2(\text{g})$ was absorbed completely. 25mL of the above solution was then treated with excess of BaCl_2 solution where all the carbonate was precipitated as $\text{BaCO}_3(\text{s})$. The solution was filtered off and the filtrate required 25mL of a 0.005M HCl solution for neutralization. Answer the following four questions based on the given observations.

3. ppm strength of $\text{CO}_2(\text{g})$, volume by volume i.e., mL of CO_2 per 10^6 mL of air was -
(A) 560 (B) 5600 (C) 100 (D) 1000
4. Weight (in milligrams) of precipitate $\text{BaCO}_3(\text{s})$ obtained from the 25 ml of test solution was
Molar masses : Ba = 137, C = 12, O = 16
(A) 27.58 (B) 275.8 (C) 4.925 (D) 49.25
5. Fraction of original NaOH (by mole) that reacted with CO_2 was -
(A) 0.2 (B) 0.4 (C) 0.6 (D) 0.8

SECTION-II : Matrix-Match Type

This Section contains **1 question**. Question has **four statements** (A, B, C and D) given in **Column I** and five statements (P, Q, R, S and T) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question, against statement B, darken the bubbles corresponding to Q and R in the ORS. 8(0)

1. Match the column-

Column-I

(Concentration of aqueous solution)

(A) 2M NaOH solution *R, P*

(B) $8\% \left(\frac{w}{V} \right)$ KOH solution *R, P*

(C) $25\% \left(\frac{w}{W} \right)$ CaCO₃ solution *Q, T*

(D) $X_{C_3H_7OH} = \frac{1}{11}$ *Q, S*

Column-II

(Density of given solutions is 1.2 g/ml)

(P) 16gm solute in 240gm solution

(Q) 60gm solute in 240 gm solution

(R) 8gm solute in 100 ml solution

(S) 30 gm solute in 100 ml solution

(T) 1 mole solute in 400 gm solution

SECTION-IV : (Integer Value Correct Type)

This section contains **5 questions**. The answer to each question is a **single digit Integer**, ranging from **0 to 9** (both inclusive) 4(-1)

1. 0.5 gm of fuming H₂SO₄ (Oleum) is diluted with water. This solution is completely neutralised by 26.7ml of 0.4 M NaOH solution. Calculate the percentage of free SO₃ in the given sample.

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

2. A container contains 1 litre, 2M solution of cyclobutane in ether. A piece of 3 kg charcoal is dipped in the solution. Molecules of cyclobutane get adsorbed on the surface of charcoal and form monolayer of cyclobutane. The molarity of resulting solution decreases to 1M. If surface area available for adsorption on charcoal 2 cm²/gm. Then find distance (in Å) between two adjacent carbon atoms in a cyclobutane molecule. (Assume: Shape of cyclobutane molecule perfect square.) [Use: N_A = 6 × 10²³]

First multiply your answer with 100, then fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

3. A solvent 'X' (mol mass 50) contains solute A (mol. mass 125) & solute 'B' (mol mass 100). If solution is 4M 'A' & 6M 'B' find simplest whole number ratio of moles of A : B : X. [Given : d_{solution} = 1.3 gm/ml].
If your answer is A : B : C, then fill your answer is A + B + C.

4. Find the sum of molarity of all the ions present in one liter of an aqueous solution of 5M NaNO₃ and 3m BeCl₂. The specific gravity of the given solution is 1.665. Assume 100 % dissociation of each salt. (Be = 9)

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

5. If 87 g of K₂SO₄ (molar mass = 174 g) is dissolved in enough water to make 250 mL of solution, calculate sum of molar concentration of [K⁺] + [SO₄²⁻].

PHYSICAL CHEMISTRY

TIME : 30 MIN.

SECTION-I : (ii) One or more options correct Type

This section contains **2 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(-1)**

1. 5 moles of VO & 6 mole of Fe_2O_3 are allowed to react completely according to reaction



The number of moles of V_2O_5 formed is -

- (A) 6 (B) 2 (C) 3 (D) 5

2. Aqueous solution containing 30gm CH_3COOH are -

- (A) 250 ml of 2M CH_3COOH solution
(B) 600 gm of 5% (by wt.) CH_3COOH solution
(C) 111gm of solution in which mole fraction of CH_3COOH is 0.1
(D) 500 gm of 7m CH_3COOH solution

(iii) Paragraph Type

This section contains **2 paragraphs** each describing theory, experiment, data etc. **6 questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Paragraph for Q.3 to Q.5

One of the major use of H_2O_2 is for artificial respiration. It is stored in cylinder of 3 litre capacity completely filled with H_2O_2 solution of '45.4V'. In a hospital of kota, patient under artificial respiration

takes 200 ml O_2 per min at STP and a cylinder last for $\frac{22.7}{8}$ hours. After that it can not be used

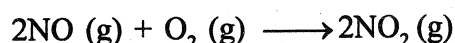
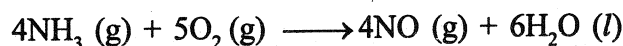
for respiration though it still contains H_2O_2 .

[Assume volume of solution and rate of decomposition remain constant.]

3. Initial moles of H_2O_2 in a cylinder are -
(A) 12 (B) 16 (C) 8 (D) 24
4. Volume of oxygen (at STP) used for respiration is
(A) 11.35 L (B) 22.7 L (C) 34.05 L (D) 17.025 L
5. Volume strength of H_2O_2 left in solution is
(A) 68.1 V (B) 22.7 V (C) 17.025 V (D) 34.05 V

Paragraph for Q.6 to Q.8

For the given series of reaction



6. If 20 ml of NH_3 is mixed with 100 ml of O_2 . Volume contraction at the completion of above reactions is
(A) 20 ml (B) 85 ml (C) 35 ml (D) 100 ml

7. To obtain maximum mass of NO_2 from a given mass of a mixture of NH_3 and O_2 , the ratio of mass of NH_3 to O_2 should be

(A) $\frac{4}{7}$ (B) $\frac{17}{56}$ (C) $\frac{17}{40}$ (D) $\frac{17}{28}$

8. Total volume of O_2 used if 20 ml NH_3 is mixed with 100 ml O_2

(A) 40 ml (B) 60 ml (C) 35 ml (D) 70 ml

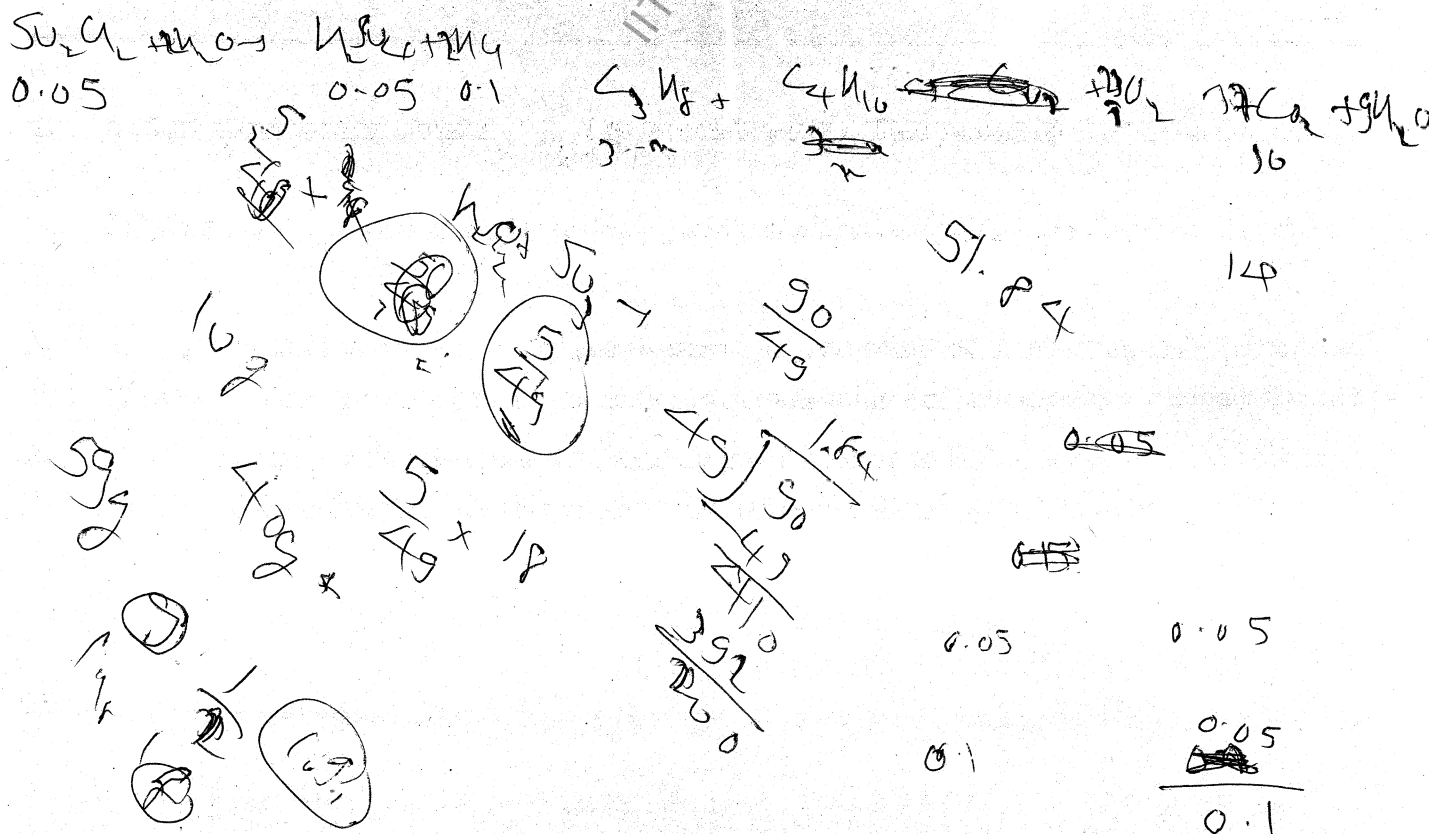
SECTION-IV : (Integer Value Correct Type)

This section contains 3 questions. The answer to each question is a single digit Integer, ranging from 0 to 9 (both inclusive) 4(-1)

1. A 50 gm oleum sample contains $\left(\frac{400}{49}\right)$ gm of combined SO_3 . Find percent label of the oleum sample.

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

2. A 3 L gas mixture of propane (C_3H_8) and butane (C_4H_{10}) on complete combustion at 25°C produced 10 L CO_2 . Assuming constant P and T conditions what was volume of butane present in initial mixture ?
3. SO_2Cl_2 (sulphuryl chloride) reacts with water to give a mixture of H_2SO_4 and HCl . What volume of 0.1M $\text{Ba}(\text{OH})_2$ (in litre) is needed to completely neutralize 50 milimole SO_2Cl_2 in water. |



SECTION-I : (i) Only One option correct Type

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(-1)**

- 0.5 gm of NaOH is required by 0.4 gm of a polybasic acid H_nA (M.W. = 96 gm) for complete neutralization. Value of 'n' and atomic weight of 'A' would be : (Assume all H atom are replaced)
☒ (A) 3, 93 (B) 2, 94 (C) 1, 95 (D) 4, 92
- Molarity and Molality of an aq. solution of a liquid (M.W. = 50) in water is 9 and 18 respectively. What is density of solution—
 (A) 1g/cc ☒ (B) 0.95 g/cc (C) 1.05 g/cc (D) 0.662 g/cc
- 300 ml, 2M H_2SO_4 solution is mixed with 200 ml, 2M $Ba(OH)_2$ solution then find final molarity of sulphate ion in diluted solution, if resulting solution is diluted to 4 times. ($BaSO_4$ formed is precipitated)
 (A) 0.2 M (B) 0.8 M (C) 0.3 M ☒ (D) 0.1 M
- What is the specific gravity of a liquid if 260 mL of the liquid has the same mass as 390 mL of water ?
 (A) 0.66 (B) 0.5 ☒ (C) 1.5 (D) 1.8

(ii) One or more options correct Type

This section contains **2 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(-1)**

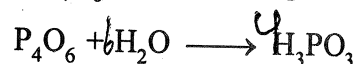
- 50 ml of CO is mixed with 20 ml of oxygen and sparked. After reaction, the mixture is treated with an aqueous KOH solution. Choose the correct option -
☒ (A) The volume of the CO that reacts is 40 ml
☒ (B) The volume of the CO_2 formed is 40 ml
 (C) The volume of the CO that remains after treatment with KOH is 20ml.
☒ (D) The volume of mixture obtained after reaction when treated with KOH solvent remains 10 ml
- Which of the following solutions will be definitely basic
☒ (A) 400 ml of 0.1 M NaOH is mixed with 100 ml of 0.4 M CH_3COOH
☒ (B) Equal volumes of same % w/v of NaOH and H_2SO_4 solutions are mixed.
 (C) Different volumes of 0.2 M NaOH and 0.1 M HCl solution are mixed. ☒
 (D) Equal masses of same % w/w NaOH and HCl solution are mixed. ☒

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This section contains **1 paragraph** each describing theory, experiment, data etc. **3 questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D).

Paragraph for Question 7 to 9

Similar to % labelling of oleum, a mixture of H_3PO_3 and P_4O_6 is labelled as $(100 + x)\%$ where x is maximum mass of water which can reacts with P_4O_6 present in 100 gm mixture of H_3PO_3 and P_4O_6



7. If such a mixture is labelled as 127%, mass of free P_4O_6 in 100 gm of mixture is -
 (A) 45 gm (B) 55 gm (C) 71 gm (D) 58 gm
8. For such a mixture of P_4O_6 and H_3PO_3 labelled as $(100 + x)\%$. Value of " x " can lie in range of -
 (A) $0 < x < 22.5$ (B) $0 < x < 49.09$ (C) $0 < x < 59.65$ (D) $0 < x < 34.66$

9. $\frac{100}{3}$ gm mixture of H_3PO_3 and P_4O_6 (labelled as 123%) is mixed with 50 L of H_2O and resulting solution (assuming volume change) is mixed with 2M NaOH solution. Volume of NaOH required for complete neutralisation is-

- (A) 250 ml (B) 750 ml (C) 200 ml (D) 500 ml

SECTION-IV : (Integer Value Correct Type)

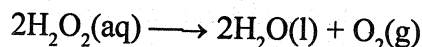
This section contains **3 questions**. The answer to each question is a **single digit Integer**, ranging from **0 to 9** (both inclusive) 4(-1)

1. An alkene upon combustion produces $\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\text{g})$. In this combustion process, if there is no volume change then the no. of C atoms per molecule of alkene, is. 2

2. 100 gm oleum sample (labelled as 107.8%) is mixed with 7.8 gm water and requires 1.1 L of x molar aq. solution of NaOH for complete neutralization. The value of x is :

3. $2\text{H}_2\text{O}_2(\text{aq}) \longrightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$ 1

Under conditions where 1 mole of gas occupies 24 dm^3 , X L of $\frac{1}{24}$ M solution of H_2O_2 produces 3 dm^3 of O_2 . Thus X is :- 6

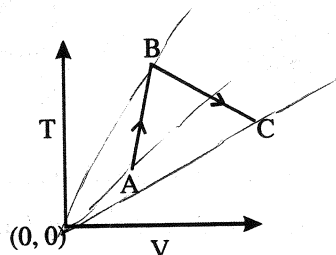


CHEMISTRY

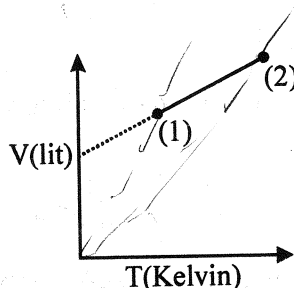
SECTION-I : (i) Only One option correct Type

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(-1)**

1. In the given isobaric process shown by graph between T & V.



- (A) Moles decreases throughout
(B) Moles first increases then decreases
(C) Moles first decreases then increases
(D) Moles cannot be predicted from given data
2. At 0.821 atm and at 177°C, gas SO_3 occupies 45L, moles of neutron present in SO_3 are (Take $N_A = 6 \times 10^{23}$)
(A) 40 (B) 24×10^{24} (C) 24×10^{23} (D) 6×10^{24}
3. A gas at a pressure of 5.0 atm is heated from 0°C to 546°C and simultaneously compressed to one-third of its original volume. Hence final pressure is
(A) 10 atm (B) 30 atm (C) 45 atm (D) 5 atm
4. For the below graph which of the following combination is correct. ('n' represent mole & 'P' represent pressure)

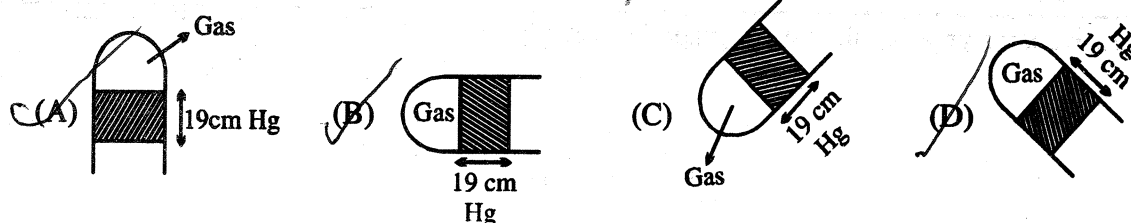


- I If 'n' is constant throughout then 'P' at 2nd point must be greater than at 1st point.
II If 'n' is doubled in moving from 1st point to 2nd point then 'P' also doubles in the process
III If 'n' is constant then 'P' is maximum at a point (1).
- (A) I is true (B) II are correct
(C) both III & II are true (D) I, II & III all are correct
5. $\text{NH}_3(\text{g})$ evolved from 1.4 gm of protein sample can be completely neutralized by 40 ml of 0.4M HCl. % (by mass) of nitrogen in the sample is -
(A) 8% (B) 16% (C) 19.4% (D) 24%

(ii) One or more options correct Type

This section contains **2 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(-1)**

6. In which of the following case(s) pressure of gas is less or equal to atmospheric pressure.



7. If the difference in the level of Hg in an open arm manometer [One end open to atmosphere and other end is connected to gas chamber] is 5 mm. Then what can be the pressure of gas ?

(Given : $1 \text{ atm} = 1.01325 \times 10^5 \text{ N/m}^2$)

- (A) 765 mm of Hg (B) 755 mm of Hg (C) 765 torr (D) 700 torr

(iii) Paragraph Type

This section contains **1 paragraphs** each describing theory, experiment, data etc. **2 questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Paragraph for Q.8 & Q.9

Two substance $A_2(g)$ and $B_2(g)$ are allowed to react to form either of $A_2B_2(l)$, $A_2B_4(g)$ or both depending on relative amount.

Answer the following questions, if 75g A_2 and 200 g B_2 are mixed.

[Given : At mass of A = 25, B = 50]

8. What is the volume of all the gases present in the mixture after completion of reaction at 760 mm of Hg and 273°C

- (A) 67.2 L (B) 44.8 L (C) 22.4 L (D) 11.2 L

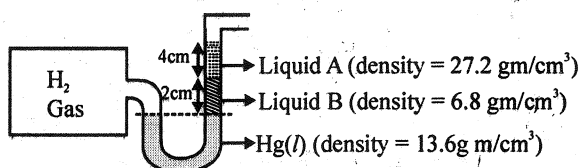
9. Now A_2B_4 is transferred to other container completely. At high temperature it is completely dissociated in its gaseous elements A_2 and B_2 . Find average molecular weight of gaseous mixture

- (A) $\frac{125}{3}$ (B) $\frac{250}{3}$ (C) 125 (D) 250

SECTION-I : (i) Only One option correct Type

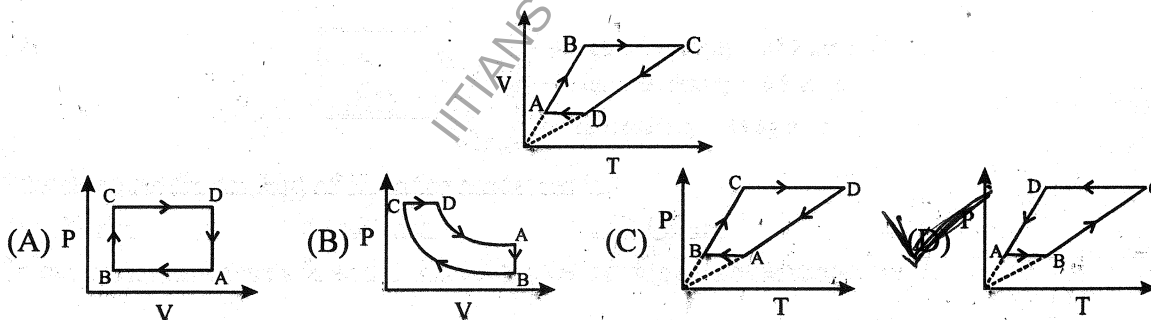
This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(-1)**

1. If all hydrogen atoms are present in its isotopic form of deuterium (${}_1\text{H}^2$) in an oleum sample. Calculate percentage by mass of sulphur in 118% of one such oleum sample ($\text{SO}_3 + \text{D}_2\text{SO}_4$)
(A) 32% (B) 64% (C) 46.8% (D) ~~37.76%~~ **37.76%**
2. A container of volume 2 litre contains H_2 gas at 300K as shown ($P_{\text{atm}} = 76 \text{ cm of Hg}$)



The pressure (in cm Hg) of H_2 in the container is -

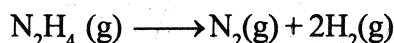
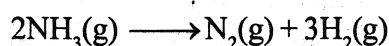
- (A) 82 cm (B) ~~85 cm~~ (C) 80 cm (D) 81 cm
3. In two different vessels X and Y, $\text{H}_2\text{O}(\text{l})$ is kept at the same temperature, the vacant space left over the surface of $\text{H}_2\text{O}(\text{l})$ is V and 3V respectively. What is the mass ratio $\frac{(W_{\text{H}_2\text{O}})_x}{(W_{\text{H}_2\text{O}})_y}$ of vapours in two vessels
(A) 1 : 3 (B) 3 : 1 (C) 1 : 4 (D) 1 : 1
 4. A cyclic process ABCD is shown in VT diagram for an ideal gas. Which of the following diagram represents the same process ?



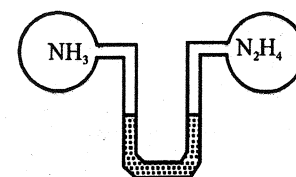
(ii) One or more options correct Type

This section contains **1 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(-1)**

5. Figure shows a manometer, one arm is connected with a bulb containing NH_3 and other arm containing N_2H_4 , both at initial pressure of 1 atm. Initially both arms of manometer contains liquid s at same level. What will be difference in level of arms, when 50% $\text{NH}_3(\text{g})$ and 75% of $\text{N}_2\text{H}_4(\text{g})$ dissociates according to the given reaction. Assuming temperature constant -



- (A) 76 cm if liquid having density 13.6 gm/ml is used
(B) 38 cm if liquid having density 27.2 gm/ml is used
(C) 152 cm if liquid having density 6.8 gm/ml is used
(D) 304 cm if liquid having density 3.4 gm/ml is used



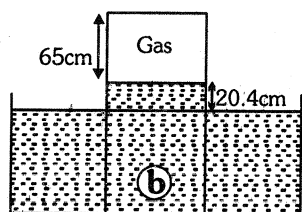
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(iii) Paragraph Type

This section contains **1 paragraphs** each describing theory, experiment, data etc. **3 questions** relate to 1 paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Paragraph for Q.6 to Q.8

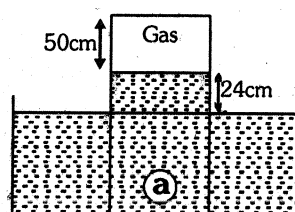
If same amount of gas is trapped over liquid **a** and liquid **b** in following containers. Assuming temperature and cross-sectional area of container are same.

$$P_{\text{atm}} = 760 \text{ mm Hg} ; d_{\text{Hg}} = 13.6 \text{ gm/mL}$$



Aqueous tension = 110 mm Hg

Density = —



Aqueous tension = 50 mm Hg

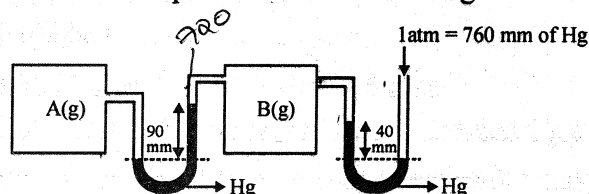
Density = 3.4 g/mL

6. Find the pressure (in cm Hg) of gas over liquid **a**.
(A) 76 cm (B) 11 cm (C) 50 cm (D) 65 cm
7. Find the pressure (in cm Hg) of gas over the liquid **b**.
(A) 76 cm (B) 11 cm (C) 50 cm (D) 65 cm
8. The density of liquid **b** will be:
(A) 5 g/cc (B) 10 g/cc (C) 15 g/cc (D) 20 g/cc

SECTION-IV : (Integer Value Correct Type)

This section contains **3 questions**. The answer to each question is a **single digit Integer**, ranging from **0 to 9** (both inclusive) **4(-1)**

1. Based on following reaction, give minimum possible value of 'x' for 1 mol AF_6 .
 $\text{AF}_6 + 3\text{H}_2\text{O} \rightarrow \text{AO}_x + 6\text{HF}$ **(3)**
2. In order to remove Pb^{2+} from 10 litre H_2O , $\text{Na}_2\text{H}_2\text{EDTA}$ (0.4 M, 100 mL) is required.
 $\text{PbCl}_2(\text{aq.}) + \text{Na}_2\text{H}_2\text{EDTA} \rightarrow 2\text{NaCl} + \text{PbH}_2\text{EDTA}$
Hence millimoles of PbCl_2 present in 1 litre of H_2O is. **(4)**
3. At 300K, two gases are filled in two equal sized containers as given.



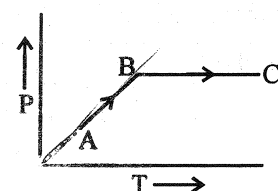
What will be the pressure of A(g) (in mm of Hg).

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

(9)

SECTION-I : (i) Only One option correct Type

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(-1)**

- 500mL, 0.2M Na_2SO_4 solution is mixed with 100 mL, 17.1% $\left(\frac{w}{V}\right)$ $\text{Al}_2(\text{SO}_4)_3$ solution and resulting solution is diluted to 5 times. Find the molarity of SO_4^{2-} ions in the final solution.
[Mw of Al = 27, s = 32, Na = 23]
(A) $\frac{1}{12}$ M (B) $\frac{5}{12}$ M (C) $\frac{7}{5}$ M (D) $\frac{12}{5}$ M
- A gas is taken isochorically from state A to state C as shown in the graph. Choose the correct statement-
(A) Moles of gas first remains constant and then increases
(B) Moles of gas first increases and then remains constant.
(C) Moles of gas first remains constant and then decreases
(D) Moles of gas first decreases and then remains constant.

- Four particles have speed 2, 3, 4 and 5 cm/s respectively. Their rms speed is :
(A) 3.5 cm/s (B) $(27/2)$ cm/s (C) $\sqrt{54}$ cm/s (D) $(\sqrt{54}/2)$ cm/s
- Human lungs can absorb 8 gm O_2 per hour by respiration. If all oxygen atoms are converted to carbohydrates ($\text{C}_6\text{H}_{12}\text{O}_6$) how long will it take to produce 180 gm $\text{C}_6\text{H}_{12}\text{O}_6$
(A) 8 hour (B) 12 hour (C) 10 hour (D) 6 hour

(iii) Paragraph Type

This section contains **2 paragraphs** each describing theory, experiment, data etc. **6 questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Paragraph for Q.5 to Q.7

Equal masses (W gram each) of three non-reacting gases X, Y and Z were mixed in sealed rigid container and total pressure at given temperature 'T' was found to be 'P' atmosphere. Now '2W' gram of X is further added to same container and temperature was raised to '2T'. At '2T' temperature, new pressure was found to be '4P' atmosphere. Now '3W' gram of Y was added further and temperature was raised to '4T'. New pressure was found to be '12P'

- Based on above information, lightest and heaviest gases are respectively -
(A) Y, Z (B) X, Z (C) X, Y (D) Z, X
- Partial pressure of X & Y after adding '2W' gram of X in container are respectively -
(A) $P, \frac{P}{3}$ (B) $2P, \frac{P}{6}$ (C) $2P, \frac{P}{3}$ (D) $3P, \frac{2P}{3}$
- In final conditions partial pressure of 'Z' is-
(A) $\frac{P}{3}$ (B) $\frac{3P}{4}$ (C) $\frac{2P}{3}$ (D) $\frac{P}{6}$

Paraphrase for Question 8 to 10

The constant motion and high velocities of gas particles lead to some important practical consequences. One such consequence is that as minimum rapidly when they come in contact. The mixing of different gases by random molecular motion and with frequent collisions is called diffusion. A similar process in which gas molecules escape through a tiny hole into a vacuum is called effusion.

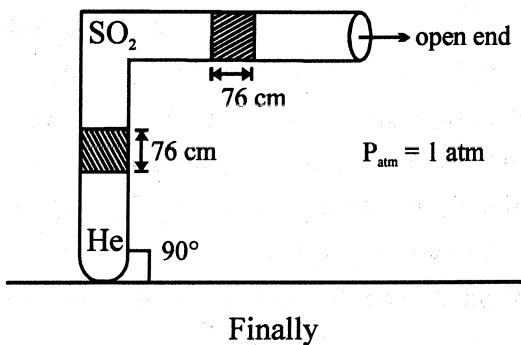
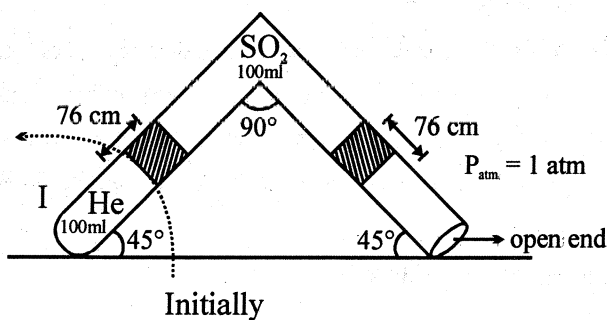
8. Helium gas at 1 atm and SO_2 at 2 atm pressure, temperature being the same, are released separately at the same moment into 1 m long evacuated tubes of equal diameters. If helium reaches the other end of the tube in t sec, what distance SO_2 would traverse in the same time interval in the other tube ?
(A) 25 cm (B) 50 cm (C) 60 cm (D) 75 cm
9. 4 g of H_2 effused through a pinhole in 10 sec at constant temperature and pressure. The amount of oxygen effused in the same time interval and at the same conditions of temperature and pressure would be :
(A) 4 g (B) 8 g (C) 16 g (D) 32 g
10. For 10 min. each at 27°C , from two identical bulbs helium and an unknown gas X at equal pressures are leaked into a common vessel of 3 L capacity. The resulting pressure is 4.1 atm and the mixture contains 0.4 mol of helium. The molar mass of gas X is :
(A) 16 (B) 32 (C) 64 (D) None of these

SECTION-IV : (Integer Value Correct Type)

This section contains 2 questions. The answer to each question is a single digit Integer, ranging from 0 to 9 (both inclusive) 4(-1)

1. A thin glass tube (uniform cross-section) is filled with He & SO_2 in between Hg column as shown in diagram. When it is rotated by angle 45° anticlockwise as shown in diagram, calculate the sum of final volume of both gases. (Assume T remains constant). [Use: $\frac{1}{\sqrt{2}} = 0.7$]

80



Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

2. An oleum sample is labelled as $(100 + x)\%$ and it contain $\frac{80}{3}\%$ free SO_3 by weight. Hence x is.

6

SECTION-I : (ii) One or more options correct Type

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct.

4(-1)

1. A gas cylinder contain 320 gm of O_2 at 30 atm and $27^\circ C$. What mass (in gram) of O_2 would escape if first the cylinder is heated to $127^\circ C$ and then valve is held open until the pressure inside the cylinder becomes 1 atm (the temperature being maintained at $127^\circ C$).
(A) 312 (B) 315 (C) 340 (D) 320
2. If two gases 'A' and B and temperature T_A and T_B respectively have identical Maxwellian plots, then which of the following statement is true -
(A) $T_B = T_A$
(B) $M_B = M_A$
(C) $\frac{T_A}{M_B} = \frac{T_B}{M_A}$
(D) Gases A and B may be O_2 and SO_2 at $27^\circ C$ and $327^\circ C$ respectively
3. Which of the following statements is not correct regarding compressibility factor of real gas?
(A) At Boyle's temperature, compressibility factor may be 1.04 at some pressures.
(B) For a gas in which only significant repulsive forces are existing, 'z' will always be greater than 1.
(C) As pressure tends to zero, compressibility factor of all real gases approaches towards unity.
(D) At Boyle's temperature, compressibility factor may be 0.96 at some pressure.
4. Select the correct statement(s):
(A) Ratio of gm/litre & % w/v of a solution is independent of solute substance.
(B) Ratio of % w/v and molarity of a solution depends on solute substance.
(C) Ratio of % w/v and molarity of a solution depends on solvent substance
(D) Ratio of % w/v & ppm for any solution is same
5. Select the incorrect statement(s)
(A) The compressibility factor (Z) for H_2 and He is given by equation : $Z = 1 + \frac{Pb}{RT}$
(B) At critical temperature, all real gases behave like an ideal gas
(C) The compressibility factor for a vanderwaals gas at critial state is $\frac{8}{3}$
(D) The r.m.s velocity of ideal gas molecule will be doubled if pressure is made four times by decreasing the volume of container
6. A bottle of oleum is labelled 109%, the maximum H_2SO_4 that can be obtained from $\left(\frac{x}{2}\right)$ gm of this oleum required 11.12 ml of 0.5M $Ca(OH)_2$ solution for complete neutralization. Find "x"
(A) 0.25 (B) 0.5 (C) 2 (D) 1

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(iv) Matching List Type

This Section contains 1 multiple choice questions. Each question has matching lists. The codes for the lists. have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **4(0)**

7. **Column-I** **Column-II**
- (P) 100 gm of 9.8 % H_2SO_4 by mass [$d_{\text{sol.}} = 1.8 \text{ g/ml}$] (1) 3.6M
- (Q) 100 gm of 9.8 % H_3PO_4 by mass [$d_{\text{sol.}} = 1.2 \text{ g/ml}$] (2) 1.2 M
- (R) $1.8 N_A$ molecules of HCl in 500 ml solution (3) 1.8 moles
($d_{\text{solution}} = 1 \text{ gm ml}^{-1}$)
- (S) On mixing 250 ml 4M NaOH and 250 ml of 1.6 M $\text{Ca}(\text{OH})_2$. Final OH^- ion. ($d_{\text{solution}} = 1 \text{ gm ml}^{-1}$) (4) 1.1m

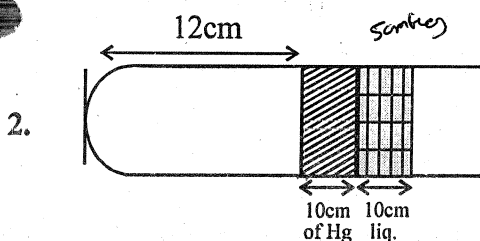
Code :

	P	Q	R	S
(A)	4	1, 2, 4	2, 3	2, 4
(B)	2, 3	2, 4	1, 2, 4	1, 4
(C)	4	2, 4	1, 3	1, 3
(D)	1, 2, 4	2, 4	1, 3	1, 4

SECTION-IV : (Integer Value Correct Type)

This section contains 3 questions. The answer to each question is a **single digit Integer**, ranging from 0 to 9 (both inclusive) **4(-1)**

1. If volume of 1M aqueous NaOH solution required to react completely with 66 gm of an aqueous acetic acid solution in which mole fraction of acid is 0.2 is $\frac{1}{x}$ litre then calculate value of x. **(2)**



$P_{\text{atm}} = 75 \text{ cm}$
 $\rho_{\text{liq.}} = 6.8 \text{ gm/cc}$

If above tube is held vertical with open end upward then find the length of air column. **(11)**

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

3. A complex compound of iron has molar mass = 2800 and it contains 8% iron by weight. The number of iron atoms in one formula unit of complex compound is : **(4)**

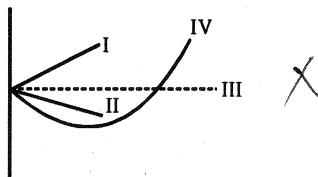
PHYSICAL CHEMISTRY

Time : 45 Min.

(ii) One or more options correct Type

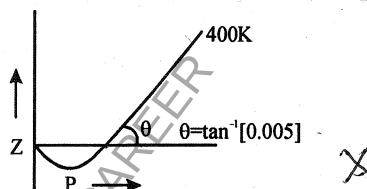
This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. 4(-1)

1. The diagram display the plot of the compression factor Z versus P for a some gases -

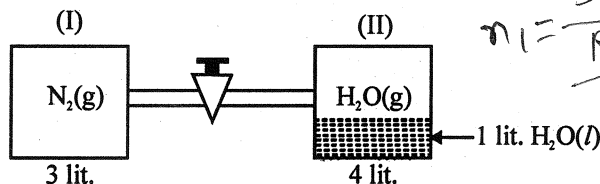


Which of the following statement(s) is/are correct for a vander wall gas -

- (A) The plot I is applicable provided the vander waal's constant 'a' is negligible $Z = 1 + \frac{bP}{RT}$
(B) The plot II is applicable provided the vander waal's constant 'b' is negligible $Z = 1 - \frac{aP}{RT^2}$
(C) The plot III is applicable provided that vander waal constant 'a' and 'b' are negligible.
(D) The plot IV is applicable provided the temperature in the gas is much higher than its Boyle's temperature
2. If critical temperature for the real gas is 500K then the value of - [$R = 0.08 \text{ l-atm/mole-K}$]



- (A) boyle's temperature is 1687.5K
(B) Vander waal constant a for the gas is $22.7 \text{ atm-L}^2/\text{mol}^2$
(C) Gas will show ideal behaviour at high temeperature & low pressure.
(D) At boyle's temperature behaviour of gas is always ideal
3. Two containers are connected by a tube of negligible volume, container (I) has $\text{N}_2(\text{g})$ gas at a pressure 4 atm and temperature T (K) and container (II) has 1 litre $\text{H}_2\text{O}(\text{l})$ at temp (T) initially. Find correct option(s) after stopcock is removed. (aq. tension = 190 torr)



- (A) Pressure of $\text{N}_2 = 2.0 \text{ atm}$ ✓
(B) Total pressure = 1.95 atm
(C) Total pressure = 2.25 atm ✓
(D) Pressure of $\text{H}_2\text{O}(\text{g})$ is 0.25 atm ✓

4. Critical constant for a real gas are given as
 $T_c = 180 \text{ K}$; $V_c = 0.123 \text{ L/mole}$
 $P_c = 45 \text{ atm}$;
The correct statement for the real gas is -

- (A) The actual volume of single gas molecule is $\left(\frac{0.123 \times 10^{-3}}{6.023 \times 10^{23}} \right) \text{ cc}$ ✓
(B) $b = 3 \times 0.123 \text{ L mol}^{-1}$ ✓
(C) The Boyle's temperature is less than 180K ✓
(D) Gas cannot be liquified at 200K

5. If volume occupied by CO_2 molecule is negligible for 1 mol of $\text{CO}_2(\text{g})$ in 2L vessel at 273K. Given for CO_2 :

$$a = \frac{3.6 \text{ atm L}^2}{\text{mole}^2}$$

Incorrect option is -

(A) $P = \frac{RT}{V} - \frac{a}{V^2}$ ✓

(B) $Z = 1 - \frac{a}{VRT}$ ✓

~~(C)~~ Gas is easily compressible in this state (than ideal gas)

~~(D)~~ For CO_2 , volume of the gas molecules is always negligible (all P & T)

6. If X is the total number of collision which a gas molecule registers with others per unit time under particular conditions, then the collision frequency of the gas containing N molecules per unit volume is

(A) X / N

(B) NX

(C) $2NX$

~~(D)~~ $NX / 2$

7. Equal moles of three real gases A, B & C are placed in three separate containers at same temperature & pressure. The value of vander waal's constant 'a' is **0.03 atm-lit²/mole, 0.1 atm-lit²/mole & 0.065 atm-lit²/mole** respectively for A, B & C gases. If radius of molecules of all three gases are same. Then find the correct relationship between volume of containers V_A, V_B & V_C is which A, B & C respectively are placed -

(A) $V_A > V_B > V_C$

(B) $V_A < V_B < V_C$

(C) $V_B > V_A > V_C$

~~(D)~~ $V_B < V_C < V_A$

8. Select the **incorrect** option(s).

(A) Collision frequency (Z_{11}) becomes $\frac{1}{4}$ times of initial if volume is doubled at constant temperature

~~(B)~~ Fraction of molecule in the range U_{mps} to $U_{mps} + 0.5\% U_{mps}$ is lesser for CH_4 than O_2 at same temperature.

~~(C)~~ Molar volume $V_m > 22.4 \text{ L}$ at 1 atm and 273 K when attractive force dominate.

~~(D)~~ The real gas behave as an ideal gas at critical point.

(iii) Paragraph Type

This section contains **1 paragraphs** each describing theory, experiment, data etc. **1 questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). 3(0)

Paragraph for Questions 9 to 10

For a non-ideal gas, the compressibility factor (Z) is defined as

$$Z = \frac{pV_m}{RT}; V_m = \text{molar volume}$$

Compressibility of an unknown gas at 600K and 1.0 atm was found to be 1.2. Also this gas was found to effuse 1.58 times slower than the pure methane gas under identical conditions.

$$[R = 1/12 \text{ L-atm/mol K}]$$

9. Density of the gas in the above mentioned experimental condition is -

(A) 0.98 gL^{-1}

~~(B)~~ 0.67 gL^{-1}

(C) 1.02 gL^{-1}

(D) 1.47 gL^{-1}

10. Molar volume of the gas in the given experimental condition is -

- (A) 40.8 L (B) 39.2 L (C) 60 L (D) 27.2 L

(iv) Matching List Type

This Section contains 1 multiple choice questions. Each question has matching lists. The codes for the lists. have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. 4(0)

11. One mole of $N_2(g)$ is taken in 1 litre empty container fitted with a movable piston at 300K. If it is heated to 1200K at constant pressure then match the change (List-II) in parameters (List-I) of gas as compared to initial state & select the correct code.

List-I (Parameter)	List-II (Change)
(P) Z_1 (number of collision made by a molecule per unit time)	(1) 1/8
(Q) Z_{11} (collision frequency)	(2) 2
(R) λ (mean free path)	(3) 1/2
(S) U_{rms} (root mean square speed)	(4) 4

(Code):

	P	Q	R	S
(A) ✓	3	1	4	2
(B)	1	2	4	3
(C)	3	4	1	2
(D)	2	1	4	3

SECTION-IV : (Integer Value Correct Type)

This section contains 2 questions. The answer to each question is a single digit Integer, ranging from 0 to 9 (both inclusive) 4(-1)

1. An unknown gas behaves ideally at 540K in low pressure region, then calculate the temperature (in K) below which it can be liquified by applying pressure

(Fill your answer as sum of digit) 7

2. A flask has 10 molecules out of which four molecules are moving at $7ms^{-1}$ and the remaining ones are moving at some speed of $X ms^{-1}$. If rms of the gas is $5 ms^{-1}$, what is 'X'?

3

SECTION-I : (ii) One or more options correct Type

This section contains 7 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. 4(-1)

- In hydrogen atom an orbit has a diameter of about 16.92 Å. What is the maximum number of electrons. That can be accommodated.
(A) 8 (B) 32 (C) 50 (D) 72
- Ratio of frequency of revolution of electron in the second excited state of He^+ and second state of hydrogen is
(A) $\frac{32}{27}$ (B) $\frac{27}{32}$ (C) $\frac{1}{54}$ (D) $\frac{27}{2}$
- An excited hydrogen atom returns to the ground state. The wavelength of emitted photon is λ . The principal quantum number of the excited state will be -

- (A) $\left[\frac{\lambda R + 1}{\lambda R} \right]^{1/2}$ (B) $[\lambda R(\lambda R + 1)]^{1/2}$ (C) $\left[\frac{\lambda R}{\lambda R - 1} \right]^{1/2}$ (D) $\left[\frac{1}{\lambda R(\lambda R + 1)} \right]^{1/2}$

- Atomic mass of elements are defined with respect to $1/12^{\text{th}}$ of mass of single atom of C-12 [present scale]. If reference is changed to $1/24^{\text{th}}$ of mass of single atom of C-12 [new scale]. Then select the correct statement(s)

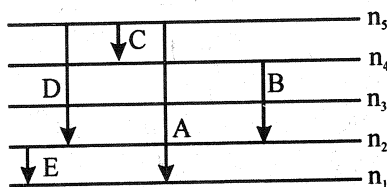
Given : Atomic mass of Fe on present scale is 56
Mass of $1/12^{\text{th}}$ of mass of single atom of C-12 = 1 amu
Mass of $1/24^{\text{th}}$ of mass of single atom of C-12 = 1 uma

- (A) Atomic mass of elements will change ✓
(B) Mass of an atom (in gm) of an element remain same in both scale ✓
(C) Mass of an atom of Fe will be 112 amu on new scale. ✗
(D) Atomic mass of Fe on new scale will be 112.

- According the vander waal's theory of non-ideal gases, which of the following production is/are correct-
(A) H_2 and He are impossible to liquify at room temperature. ✓
(B) For NH_3 , Z decreases more rapidly on increasing pressure than the methane in lower pressure region ✓

- (C) In the lower pressure region, Z decreases on increasing pressure, H_2 and He are exception to that.
(D) In the higher pressure region, Z increases with increasing pressure, H_2 and He are exception to that

- For an H-like atom which follows Bohr's model, some spectral lines were observed as shown. If it is known that line 'E' belongs to visible region, then the correct statement(s) is /are for following transition.



- (A) There cannot be any line in ultraviolet region
(B) Only line possibly belonging to ultraviolet region is A
(C) Line having shortest wavelength is A
(D) Line having least energy is C

7. In Bohr's model of the hydrogen atom-
- (A) The radius of the n^{th} orbit is inversely proportional to n^2
 - (B) The total energy of the electron in the n^{th} orbit is inversely proportional to n
 - (C) The angular momentum of the electron in an orbit is an integral multiple of $h/2\pi$
 - (D) The magnitude of potential energy of the electron in any orbit is greater than its K.E.

(iii) Paragraph Type

This section contains **2 paragraphs** each describing theory, experiment, data etc. **5 questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Paragraph for Q.8 & Q.9

In an atom when an electron jumps from higher energy level to lower energy level it emits energy in form of electromagnetic radiations. When these electromagnetic radiations are passed through a prism and received on a photographic film some lines are observed on that film and those lines are called spectral lines.

For hydrogen like species when jump takes from any excited state to ground state ($n = 1$), line produced in that case is called a lyman series line.

If transition occurs from 3rd or above levels to second level then corresponding lines produced are called Balmer lines.

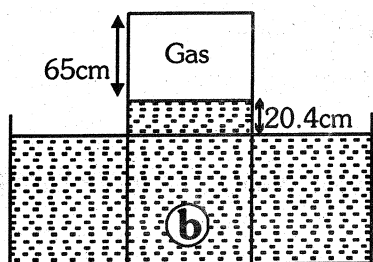
Similarly for next level it is called paschen series line.

8. The wavelength of photon corresponding to second least energy of lyman series in H-atom.
- (A) 102 nm
 - (B) 121 nm
 - (C) 91 nm
 - (D) 486 nm
9. If there are 3 atoms of a Hydrogen like species, one in 2^{nd} , one in 3^{rd} and one in 4^{th} excited state respectively then how many maximum total different Balmer and paschen lines can be produced.
- (A) 2
 - (B) 3
 - (C) 4
 - (D) 5

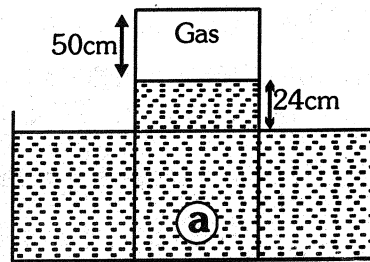
Paragraph for Q.10 to Q.11

If same amount of gas is trapped over liquid @ and liquid ① in following containers. Assuming temperature and cross-sectional area of container are same.

$P_{\text{atm}} = 760 \text{ mm Hg}$; $d_{\text{Hg}} = 13.6 \text{ gm/mL}$



Aqueous tension = 110 mm Hg
Density = —



Aqueous tension = 50 mm Hg
Density = 3.4 g/mL

10. Find the pressure (in cm Hg) of gas over liquid (a).
(A) 76 cm (B) 11 cm (C) 50 cm (D) 65 cm
11. Find the pressure (in cm Hg) of gas over the liquid (b).
(A) 76 cm (B) 11 cm (C) 50 cm (D) 65 cm
12. The density of liquid (b) will be :
(A) 5 g/cc (B) 10 g/cc (C) 15 g/cc (D) 20 g/cc

(iv) Matching List Type

This Section contains 1 multiple choice questions. Each question has matching lists. The codes for the lists. have choices (A), (B), (C) and (D) out of which ONLY ONE is correct. 4(0)

13. Column-II gives the values of vander waal's constant 'a' of a few gases which are shown in column-I. Identify the gas the with the corresponding 'a' value -

Column-I	Column-II
(P) H ₂	(A) 137.8 kPa dm ⁶ mol ⁻²
(Q) He	(B) 21.8 kPa dm ⁶ mol ⁻²
(R) O ₂	(C) 364.0 kPa dm ⁶ mol ⁻²
(S) CO ₂	(D) 3.5 kPa dm ⁶ mol ⁻²

(Code) :

	P	Q	R	S
(A)	2	4	1	3
(B)	1	2	4	3
(C)	3	4	1	2
(D)	2	1	4	3

SECTION-IV : (Integer Value Correct Type)

This section contains 1 questions. The answer to each question is a single digit Integer, ranging from 0 to 9 (both inclusive) 4(-1)

1. Two flask of equal volume are connected by a narrow tube (of negligible volume) are at 27°C and contain 0.7 moles of H₂ at 0.5 atm. One of the flask is then immersed into a bath kept at 127°C, while other remains at 27°C. If number of moles of H₂ in flask 1 and flask 2 are a and b respectively, then calculate value of 10 × |a - b|.

3
50
20.4 × 10

7

300 K



SECTION-I : (ii) One or more options correct Type

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. 4(-1)

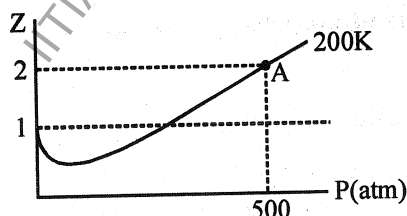
1. Which of the following statement (s) is/are true -

- (A) If H-atoms are in 3rd energy level, then number of maximum different spectral lines produced by 2 atoms will be equal to number of maximum spectral lines produced by 3 H-atoms.
(B) For third excited state, maximum number of different spectral line produced by 4 H-atoms & 4 He⁺ ions are equal.
(C) Energy evolved from 3rd to 2nd energy level transition in He⁺ can be used to ionise Li²⁺ from ground level
(D) Energy evolved by a He⁺ ion which is in its 1st excited state can ionise a hydrogen atom which is in its ground state.

2. Which of the following statement is wrong for H-like atom -

- (A) Magnitude of energy of an orbit in H-like species is directly proportional to $1/n^2$
(B) Frequency of revolution of an electron in an orbit is proportional to n^3
(C) Radius of a Bohr orbit is proportional to n^2
(D) Time period of revolution of an electron is proportional to n^3

3. 'Z' v/s 'P' graph is plotted for 1 mole of hypothetical gas. Volume of gas at the point 'A' is -
[R = 0.08 atm L/mole-K]



- (A) 0.01 L (B) 0.09 L (C) 0.064 L (D) 0.64 L

4. The angular momentum of electron in an excited H atom is $\frac{h}{\pi}$. The P.E. of electron will be :

- (A) 6.8 eV (B) 3.4 eV (C) -6.8 eV (D) -3.4 eV

5.

Gas	H ₂	He	O ₂	N ₂
Critical temp. in K	33.2	5.3	154.3	126

from the above data what would be the order of liquification of these gases ? Start writing the order for the gas which liquify first -

- (A) H₂, He, O₂, N₂ (B) H₂, O₂, H₂, N₂
(C) N₂, O₂, He, H₂ (D) O₂, N₂, H₂, He

(iii) Paragraph Type

This section contains **2 paragraphs** each describing theory, experiment, data etc. **Four questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D).

Paragraph for Q.06 & Q.07

From hydrogen gas discharge tube, light is emitted. Emitted light is used to emit electron from sodium metal of work function $\phi = 1.82$ eV.

6. If the fastest photoelectron has KE of 0.73 eV, the wavelength of incident radiation on the sodium metal is -
 (A) 5627.63 Å (B) 4862.74 Å (C) 3384 Å (D) 9872 Å
7. The transition in hydrogen atom takes place from -
 (A) $n_1 = 4$ to $n_2 = 1$ (B) $n_1 = 4$ to $n_2 = 3$ (C) $n_1 = 3$ to $n_2 = 1$ (D) $n_1 = 4$ to $n_2 = 2$

Paragraph for Q.08 & 09

In a H-like species there are two energy levels A and B above the ground state having principal quantum numbers of n_1 and n_2 respectively. A sample of this H-like species has all atoms/ions in excited levels A or B only and none in any other energy level. Energy of level B is greater than that of level A and a total of 15 different lines are emitted from this sample on returning to ground state out of which 6 lines are emitted due to electronic transitions between the levels n_1 and n_2 only. Also energy difference between levels n_2 and n_1 , $E_{n_2} - E_{n_1} = 4.53$ eV.

8. The correct option is -
 (A) $n_2 = 2, n_1 = 5$ (B) $n_2 = 6, n_1 = 3$ (C) $n_2 = 5, n_1 = 2$ (D) $n_2 = 3, n_1 = 6$
9. Calculate minimum wavelength in the above transition -
 (A) $\frac{4}{3R_H}$ (B) $\frac{9}{8R_H}$ (C) $\frac{900}{11R_H}$ (D) $\frac{36}{35R_H}$

SECTION-IV : (Integer Value Correct Type)

This section contains **2 questions**. The answer to each question is a **single digit Integer**, ranging from **0 to 9** (both inclusive) 4(-1)

1. Wavelength of radio waves is 124 nm. Total number of photons per second produced by a source that consumes energy at the rate of 16W is $x \times 10^{19}$. The value of x is - (1)
2. The ionisation potential of a hydrogen like species is 36 volt. What is the value of excitation energy from ground state to 1st excited state. (in eV)

(Fill your answer as sum of digit)

Handwritten calculations:

$$x_6 = \frac{16 \times 10^3}{124 \times 10^{-9}} = \frac{16 \times 10^{12}}{124}$$

$$x = \frac{16 \times 10^{12}}{124} = 1.29 \times 10^{11}$$

$$x = 1.29 \times 10^{11} = 129 \times 10^9$$

$$x = 129 \times 10^9 = 129 \times 10^9$$

$$36 \left\{ \frac{1}{1} - \frac{1}{4} \right\} = 36 \times \frac{3}{4} = 27$$

SECTION-I : (ii) One or more options correct Type

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. 4(-1)

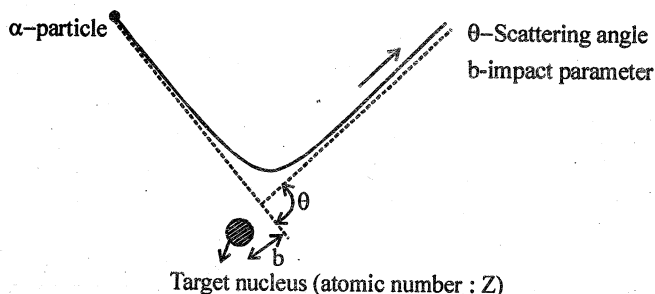
- Which statement is not true, regarding 2s orbital.
(A) Number of radial nodes is greater than zero
(B) Angular nodes is equal to zero
(C) $\Psi(\theta, \phi) = \text{constant}$
(D) Probability density is zero at nucleus
- For an electron, with $n = 3$ has only one radial node. The orbital angular momentum of the electron will be -
(A) 0 (B) $\sqrt{6} \frac{h}{2\pi}$ ~~(C) $\sqrt{2} \frac{h}{2\pi}$~~ (D) $3 \left(\frac{h}{2\pi} \right)$
- 10 litre container contains He at 210 atm and 300 K. Given value of 'b' for 'He' is $0.08 \text{ dm}^3 \text{ mol}^{-1}$. Compressibility factor is ($R = 0.08 \text{ atm lit mol}^{-1} \text{ K}^{-1}$)
(A) 1.4 ~~(B) 1.7~~ (C) 0.7 (D) 0.4
- When 3 moles of A and 1 mole of B are mixed in 1 litre vessel the following reaction takes place $A(g) + B(g) \rightarrow 2C(g)$. 1.5 moles of C are formed. The equilibrium constant for the reaction is -
(A) 0.12 (B) 0.25 (C) 0.50 (D) 4.0
- $\Psi_{(r)} = ke^{-r/K'} \cdot r^2 \cdot (r^2 - K''r + K''')$, if above orbital has no nodal plane then orbital can be :
(A) $5d_{z^2}$ (B) $5d_{xy}$ (C) $4d_{z^2}$ ~~(D) $4d_{xy}$~~
- Which is not one of the postulates of the kinetic molecular theory
(A) At a constant temp. all of the particle have the same speed
(B) There are no force of attraction between molecules
(C) Gas particles move in a straight line between collisions
(D) The molecules are in a state of constant random motion
- A particle moving with a velocity of $6.626 \times 10^7 \text{ m/s}$ has a De-Broglie wavelength of $1 \text{ \AA}$ in a circular path of radius $0.529 \text{ \AA}$. The angular momentum of particle is.
($h = 6.626 \times 10^{-34} \text{ J} \times \text{sec}$).
~~(A) $3.5 \times 10^{-34} \text{ kg m}^2 \text{ sec}^{-1}$~~ (B) $3.5 \times 10^{-35} \text{ kg m}^2 \text{ sec}^{-1}$
(C) $3.5 \times 10^{-31} \text{ kg m}^2 \text{ sec}^{-1}$ (D) $1.053 \times 10^{-34} \text{ kg m}^2 \text{ sec}^{-1}$
- Which of the following statements (regarding an atom of H) are correct ?
~~(A) kinetic energy of the electron is maximum in the first orbit~~
(B) potential energy of the electron is maximum in the first orbit
~~(C) radius of the second orbit is four times the first orbit~~
~~(D) various energy levels are equally spaced on energy scale~~

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This section contains **1 paragraph** each describing theory, experiment, data etc. **2 questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Paragraph for Q.9 to Q.10

The following text gives a qualitative idea about Rutherford's α -particle scattering experiment



Applying some trigonometry and calculus, Rutherford got the following equation -

$$\cot\left(\frac{\theta}{2}\right) = \frac{K \times (\text{K.E.})}{Ze^2} \times b$$

where $K = \text{constant} = \left(\frac{1}{9 \times 10^9}\right) \frac{C^2}{N-m^2}$

K.E. = initial kinetic energy of α -particle

$e = 1.6 \times 10^{-19} C$

9. Identify the incorrect statement **(A.)**

- ☒ (A) Scattering angle increases as impact parameter increases
- ☒ (B) Scattering angle decreases as impact parameter increases
- (C) As the initial kinetic energy of α -particle increases, scattering angle decreases
- (D) Keeping all the other factors same, the scattering angle obtained from gold nucleus is larger than the scattering angle obtained from copper nucleus

10. In an experiment with gold nucleus (assume $z = 80$) with impact parameter equal to $10^{-14} m$ the scattering angle was found to be 90° , what should be the initial K.E. of α -particle.

- ☒ (A) $1.84 \times 10^{-12} J$ (B) $184 \times 10^{-12} J$ (C) $1.44 \times 10^{-12} J$ (D) $14.4 \times 10^{-12} J$

SECTION-IV : (Integer Value Correct Type)

This section contains **2 questions**. The answer to each question is a **single digit Integer**, ranging from **0 to 9 (both inclusive)** **4(-1)**

1. In a vessel O_2 gas molecules were adsorbed on solid surface causing its partial pressure to decrease from 1 atm to 0.5 atm.

Given : $N_A = 6 \times 10^{23}$

Volume of vessel = 2.24 L

Temperature = 273 K

Total area of solid surface = 10^2 cm^2

Number of active sites per unit area = 10^{24} m^{-2}

Find number of O_2 molecules adsorbed per active site.

2. The density of a mixture of O_2 and N_2 gases at 1 atm and 273K is 0.0013 gm/ml. If partial pressure of O_2 in the mixture is A then calculate value of 25 A.

CHEMISTRY

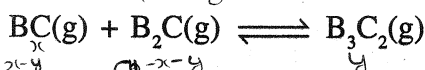
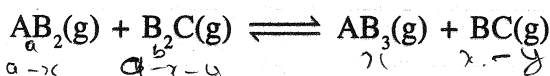
SECTION-I : (ii) One or more options correct Type

This section contains **7 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(-1)**

1. N_2 and O_2 combine at a given temperature to produce NO. At equilibrium the yield of NO is 'x' percent by volume. If $x = \sqrt{Ka.b} - \frac{K(a+b)}{4}$, where K is the equilibrium constant of the given reaction at the given temperature and a and b are initial volume percentage of N_2 and O_2 respectively. Report the maximum value of K at which x is maximum.

(A) $K = 1$ (B) $K = 2$ (C) $K = 4$ (D) $K = 16$

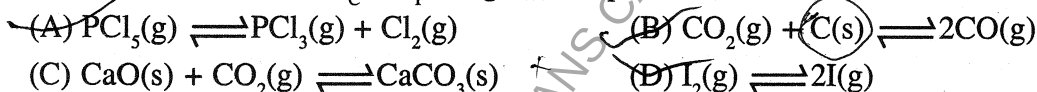
2. If two gases AB_2 and B_2C are mixed, following equilibrium are readily established.



If the reaction is started only with AB_2 and B_2C , then which of the following is nearly true at equilibrium, considering initial moles to be same.

(A) $[AB_3]_{eq} = [BC]_{eq}$ (B) $[AB_2]_{eq} = [B_2C]_{eq}$
(C) $[AB_3]_{eq} > [B_3C_2]_{eq}$ (D) $[AB_3]_{eq} < [BC]_{eq}$

3. The reaction for which $K_c < K_p$ at a given temperature is/are -



4. For the equilibrium : $NH_2COONH_4(s) \rightleftharpoons N_2(g) + 3H_2(g) + CO(g) + \frac{1}{2}O_2(g)$

If the equilibrium pressure is 22 bar. Then the value of K_p° will be [1 atm = 1.01325 bar]

(A) $\frac{11^4}{8}$ (B) $\frac{11^4}{8} \times (1.01325)^{-4}$ (C) 64
(D) $4^5 3^3 \sqrt{2}$

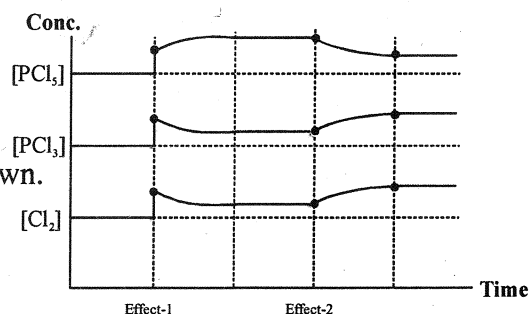
Consider given endothermic reaction at equilibrium,



A graph is plotted between concentration and time as shown.

Effect-1 & Effect-2 are due to respectively

- (A) P increase, T increase
(B) P increase, T decrease
(C) Inert gas added at constant pressure, T increase
(D) P decrease, T decrease



6. Number of N_2 molecules present in 1 litres of vessel at 1 atm, 273 K when compressibility factor is 1.2.

(A) 2.23×10^{24} (B) 2.23×10^{22} (C) 2.7×10^{22} (D) 2.7×10^{24}

7. For second reaction, if energy of activation of forward & backward reaction are respectively 85 and 42 kJ/mole, then at 400K K_c for second reaction is -

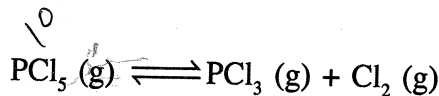
(A) 10 (B) < 10
(C) > 10 (D) can be greater or less than 10

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Paragraph Type

This section contains **1 paragraph** each describing theory, experiment, data etc. **2 questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Paragraph for Question 8 to 10

10 moles of pure PCl_5 gas is put into a closed container of volume 'V' and temperature 'T' allowed to come at equilibrium, at an equilibrium pressure 20 atm. The pure PCl_5 is found to be 50% dissociated at equilibrium.

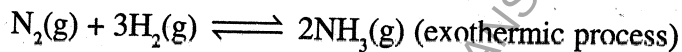


8. K_p for the above the reaction is -
(A) 20/3 (B) 100 (C) 10/3 (D) 20/6
9. The partial pressure of PCl_5 , if equilibrium is established at new equilibrium pressure 35 atm by changing volume is -
(A) 10 atm (B) 15 atm (C) 18 atm (D) 12 atm
10. If 20 moles of $\text{PCl}_5(\text{g})$ is added to original equilibrium mixture keeping total pressure constant to 20 atm at same temperature 'T', then ratio of new equilibrium volume to the initial volume V will be -
(A) 1.5 (B) 2.0 (C) 1.8 (D) 3.0

SECTION-IV : (Integer Value Correct Type)

This section contains **2 questions**. The answer to each question is a **single digit Integer**, ranging from **0 to 9** (both inclusive) **4(-1)**

1. Consider the reaction



How many of given changes will result in decrease in number of moles of N_2 gas initially present at equilibrium in a closed vessel.

- (1) increase in temperature. ~~X~~
- (2) decrease in volume of container. ~~X~~ ✓
- (3) increase in total pressure. ✓
- (4) addition of He gas at constant pressure. ~~X~~ X
- (5) addition of He gas at constant volume. X
- (6) addition of small amount of $\text{H}_2\text{O}(\text{l})$ in vessel. X
- (7) addition of H_2 gas in vessel at constant volume. X ✓
- (8) removing some mole of NH_3 . X ✓
- (9) addition of N_2 gas. X
- (10) decrease in total pressure. X

2. One litre gaseous mixture is effused in 4.5 minutes and 30 seconds while 1 litre of oxygen takes 10 minutes for effusion. The gaseous mixture contains in it ethane and hydrogen. Calculate vapour density of gaseous mixture.

PHYSICAL CHEMISTRY

TIME : 45 MIN

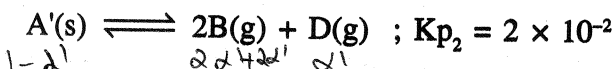
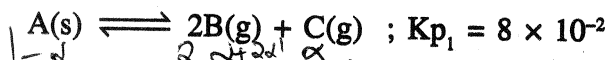
CHEMISTRY

SECTION-I : (ii) One or more options correct Type

This section contains 7 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which ONE or MORE are correct.

4(-1)

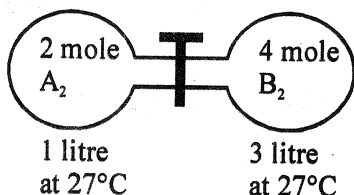
- For the balanced redox reaction
 $a\text{NO}_3^- + b\text{Cu}_2\text{O} + c\text{H}^+ \rightarrow d\text{NO} + e\text{Cu}^{2+}$
 where a, b, c, d & e are stoichiometric coefficients if c is 14 then the value of 'e' is -
 (A) 2 (B) 3 (C) 6 (D) 7
- The K_p for following reaction is $2.25 \times 10^{-2} \text{ atm}^2$.
 $\text{NH}_4\text{CN(s)} \rightleftharpoons \text{NH}_3\text{(g)} + \text{HCN(g)}$
 The value of equilibrium pressure (in atm) above $\text{NH}_4\text{CN(s)}$ is
 (A) 0.15 (B) 0.3 (C) 15 (D) 30
- For given two equilibrium attained in a container which are correct if degree of dissociation of A and A' are α and α' respectively (P' = Partial pressure at equilibrium)



- (A) $\frac{K_{p2}}{K_{p1}} = \left(\frac{3\alpha' + 2\alpha}{3\alpha + 2\alpha'} \right)^3 \frac{\alpha}{\alpha'}$ (B) $\frac{P'_C}{P_D} = 4$
 (C) $P'_B = P'_C + P'_D$ (D) $\alpha < \alpha'$

- Calculate K_c for the reaction : $\text{KI (aq.)} + \text{I}_2\text{(s)} \rightleftharpoons \text{KI}_3\text{(aq.)}$. Given that initial weight of KI is 1.326 gm and weight of KI_3 is 0.105 gm and number of moles of free I_2 is 0.0025 at equilibrium. The volume of solution is 1 litres. (M_w of KI = 166, M_w of KI_3 = 420)
 (A) 0.032 (B) 0.024 (C) 0.064 (D) 0.012

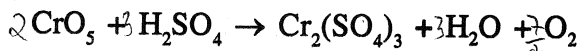
5.



The gas A_2 in the left flask allowed to react with gas B_2 present in right flask as $\text{A}_2\text{(g)} + \text{B}_2\text{(g)} \rightleftharpoons 2\text{AB(g)}$
 $K_c = 4$ at 27°C . What is the concentration of AB when equilibrium is established :-

- (A) 1.33 M (B) 0.66 M (C) 0.33 M (D) 2.66 M

- Consider the following reaction :

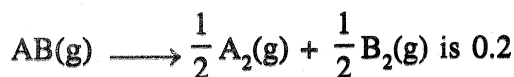


One mole of CrO_5 will librate how many moles of O_2

- (A) $\frac{7}{4}$ (B) $\frac{5}{4}$ (C) 1 (D) None of these

7. Select correct statements:

- (A) Low pressure is favourable for evaporation of $H_2O(l)$
(B) The degree of dissociation of $CaCO_3(s)$ decreased with increase in pressure
(C) If the equilibrium constant for $A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$ is 25, then equilibrium constant for



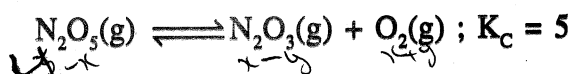
- (D) If solid product is added to an equilibrium mixture, the equilibrium will be unaffected.

(iii) Paragraph Type

This section contains **1 paragraphs** each describing theory, experiment, data etc. **2 questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Paragraph for Q.8 to Q.9

Consider following reaction at 300K-



8. If 4 moles of N_2O_5 is kept in 1 L container to attain equilibrium and at equilibrium 5 moles of O_2 are produced, then at equilibrium moles of N_2O_3 will be -
(A) 1 (B) 3 (C) 2 (D) 4
9. For second reaction, if energy of activation of forward & backward reaction are respectively 85 and 42 kJ/mole, then at 400K K_c for second reaction is -
(A) 10 (B) < 10
(C) > 10 (D) can be greater or less than 10

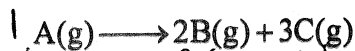
SECTION-IV : (Integer Value Correct Type)

This section contains **3 questions**. The answer to each question is a **single digit Integer**, ranging from **0 to 9 (both inclusive)** **4(-1)**

1. For a real gas, if at critical conditions molar volume of gas is 8.21 litre at 3 atm, then critical temperature (in K) will be :

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

2. Gas A taken in a closed rigid container is allowed to decompose partially to the reaction.



- The gaseous mixture formed effuses 1.5 times faster than a gas having molecular weight 105 under similar condition. Find the mole fraction of C in the gaseous mixture formed.

Given : Mol. wt. of A = 140

Mol. wt. of B = 64

Mol. wt. of C = 4

Write your answer by multiplying with 10.

3. $3A(g) \rightleftharpoons 2B(g)$ $K_p = 2 \text{ bar}^{-1}$

At what total pressure, the A is 75% dissociated

SECTION-I : (ii) One or more options correct Type

This section contains **7 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. 4(-1)

1. Consider the K_p values of some of the reaction

Reaction (A) $K_{p1} = 8 \text{ atm}$

Reaction (B) $K_{p2} = 25 \text{ atm}^{-2}$

Reaction (C) $K_{p3} = 0.8$

Reaction (D) $K_{p4} = 7.4 \text{ atm}^{+2}$

For which reaction $K_p \geq K_c$ at 300K

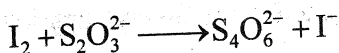
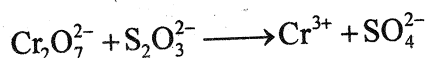
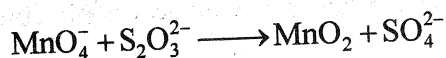
(A) Reaction A and reaction B

(B) Reaction A and reaction D

(C) Reaction B and reaction C

(D) All

2. Three different solutions of oxidising agents KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$ and I_2 is titrated separately with 0.158 gm of $\text{Na}_2\text{S}_2\text{O}_3$. If molarity of each oxidising agent is 0.1 M and reactions are -



Then ,

(A) Volume of KMnO_4 used is maximum

(B) Volume of iodine used is minimum

(C) Wt. of I_2 used in titration is maximum

(D) Gram equivalent of $\text{Na}_2\text{S}_2\text{O}_3$ are same in all the reactions.

3. Choose the **incorrect** statement(s)-

(A) 1 mole of MnO_4^- ion can oxidized 10 moles of Fe^{2+} ion in acidic medium

(B) 1 mole of $\text{Cr}_2\text{O}_7^{2-}$ ion can oxidized 12 moles of Fe^{2+} ion in acidic medium

(C) 2 mole of Cu_2S can be oxidized by 2.6 moles of MnO_4^- ion in acidic medium ($\text{Cu}_2\text{S} \rightarrow \text{Cu}^{2+} + \text{SO}_2$)

(D) 2 mole of Cu_2S can be oxidized by $8/3$ moles of $\text{Cr}_2\text{O}_7^{2-}$ ion in acidic medium ($\text{Cu}_2\text{S} \rightarrow \text{Cu}^{2+} + \text{SO}_2$)

4. Monochromatic radiations of wavelength λ are incident on a hydrogen sample in ground state. Hydrogen atom absorb the light and subsequently emits radiations of 10 different wavelengths. The value of λ is nearly-

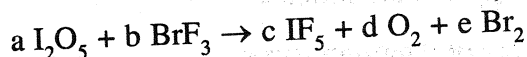
(A) 203 nm

(B) 95 nm

(C) 80 nm

(D) 73 nm

5. Consider the balanced chemical reaction :



Calculate the value of $(b + c + e)/a$.

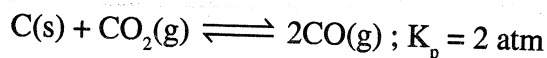
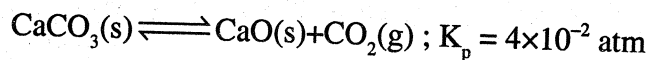
(A) 10

(B) 7

(C) 6

(D) 3

6. At 1000K, solid carbon, CO & Na_2CO_3 are mixed and allowed to attain following equilibrium, what is the pressure of CO at equilibrium (in atm)



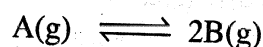
- (A) 0.04 atm (B) $5\sqrt{2}$ (C) 50 atm (D) $\frac{\sqrt{2}}{5}$ atm

7. A mixture containing substance 'X' & 'Y' is mixed with a substance 'Z'. If 'Z' reacts with both X & Y then. Select the correct option(s)

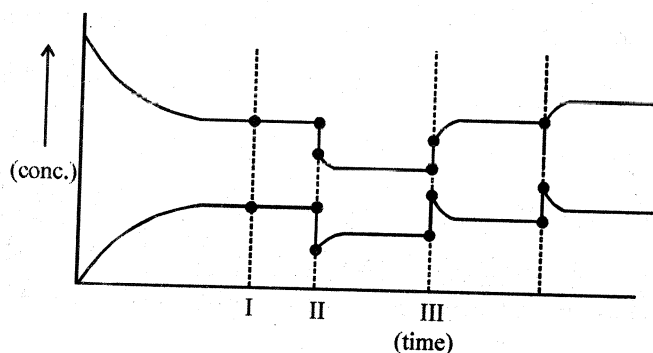
- (A) Moles of X + moles of Y = Total moles of Z reacted "is always applicable"
(B) Equivalents of X + equivalents of Y = Total equivalents of Z reacted "is always applicable"
(C) $\frac{\text{Equivalents. of X}}{n\text{-factor of Z in reaction with X}} + \frac{\text{Equivalents of Y}}{n\text{-factor of Z in reaction with Y}} = \text{Total moles of Z reacted ; is always applicable}$

- (D) $\frac{\text{Moles of Z reacted in reaction with X}}{\text{Moles of Z reacted in reaction with Y}} = \text{Total moles of Z reacted ; is always applicable}$

8. Consider the following equilibrium



and the following graph :



Select the correct statements :

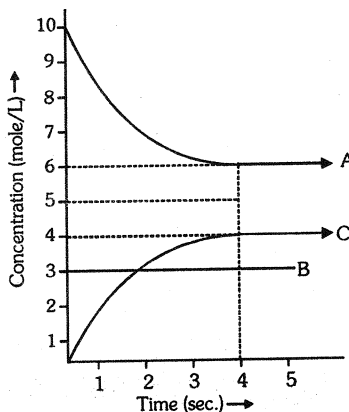
- (A) Effect (I) represent the insertion of inert gas at constant volume
(B) Effect (II) represent the insertion of inert gas at constant pressure
(C) Effect (III) represent the decrease in volume of container
(D) Effect (IV) represent the addition of moles of gas 'B'

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(iii) Paragraph Type

This section contains **1 paragraph** each describing theory, experiment, data etc. **2 questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Paragraph for Question 9 & 10

A, B and C react in the 1 : 1 : 1 stoichiometric ratio. The concentration of A, B and C were found to vary with time as shown in the fig. below :



9. Which of the following equilibrium reaction represents the correct variation of concentration with time. **[3]**
- (A) $C(g) + A(g) \rightleftharpoons B(g)$ (B) $A(g) + B(g) \rightleftharpoons C(g)$
(C) $C(g) + B(s) \rightleftharpoons A(g)$ (D) $A(g) + B(s) \rightleftharpoons C(g)$
10. The value of equilibrium constant (K_c) for the equilibrium represented in above sketch will be : **[3]**
- (A) $\frac{12}{7}$ (B) $\frac{2}{3}$ (C) 2 (D) $\frac{2}{9}$

SECTION-IV : (Integer Value Correct Type)

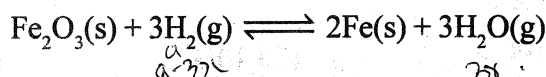
This section contains **1 questions**. The answer to each question is a **single digit Integer**, ranging from **0 to 9** (both inclusive) **4(-1)**

1. By a sample of ground state atomic hydrogen, UV light of energy $\frac{13.6 \times 48}{49} \frac{\text{eV}}{\text{quanta}}$ is absorbed. How many different wavelengths will be observed in Balmer region of hydrogen spectrum ?

SECTION-I : (i) Only One option correct Type

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. 3(-1)

- In a gaseous reaction at equilibrium, 'n' mole of reactant "A" decompose to give 1 mole each of C and D. It has been found that degree of dissociation of A at equilibrium is independent of total pressure. Value of 'n' is -
(A) 1 (B) 3 (C) 0 (D) 2
- $\text{Fe}_2\text{O}_3(\text{s})$ may be converted to Fe by reaction :

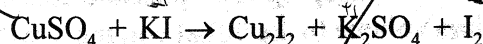


For which $K_c = 8$ at temperature 800K.

What percentage of H_2 remains unreacted till equilibrium

- (A) 50% (B) 66.6% (C) 33.3% (D) 78%

- The n_{factor} of KI in the following reaction is-



- (A) 1 (B) 2 (C) 1/4 (D) 1/2

- In the reaction $\text{Ca} + \text{H}_2 \rightarrow \text{CaH}_2$

Select the incorrect statement -

- (A) Calcium undergoes oxidation (B) Hydrogen undergoes reduction
(C) Calcium act as oxidising agent (D) Hydrogen act as oxidising agent

- Which is not one of the postulates of the kinetic molecular theory

- (A) At a constant temp. all of the particle have the same speed
(B) There are no force of attraction between molecules
(C) Gas particles move in a straight line between collisions
(D) The molecules are in a state of constant random motion

- 1 mole of $\text{H}_2\text{C}_2\text{O}_4$ is oxidised by X mole of MnO_4^- in strong basic medium and 1 mole of NaHC_2O_4 is oxidised by y mole of MnO_4^- in acidic medium. Ratio of x/y is-

- (A) 2 : 1 (B) 5 : 1 (C) 3 : 1 (D) 1 : 3

- X gm of metal gave 'Y' gm of its oxide so equivalent mass of metal is :

- (A) $\left(\frac{X}{Y-X}\right) \times 8$ (B) $\left(\frac{Y-X}{X}\right) \times 8$ (C) $\left(\frac{Y+X}{X}\right) \times 8$ (D) $\frac{X}{Y} \times 8$

- 25 ml of $\frac{N}{10}$ caustic soda solution exactly neutralises 20 ml of an acid solution contained 7.875 gm of acid per liter. What will be equivalent mass of acid.

- (A) 63 (B) 126 (C) 26 (D) 25

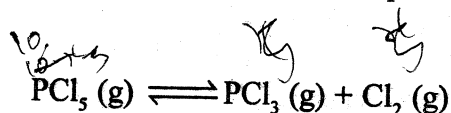
(iii) Paragraph Type

This section contains **1 paragraphs** each describing theory, experiment, data etc. **3 questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D).

3(0)

Paragraph for Question 9 to 11

10 moles of pure PCl_5 gas is put into a closed container of volume 'V' and temperature 'T' allowed to come at equilibrium, at an equilibrium pressure 20 atm. The pure PCl_5 is found to be 50% dissociated at equilibrium.



9. K_p for the above the reaction is -
 (A) 20/3 (B) 100 (C) 10/3 (D) 20/6
10. The partial pressure of PCl_5 , if equilibrium is established at new equilibrium pressure 35 atm by changing volume is -
 (A) 10 atm (B) 15 atm (C) 18 atm (D) 12 atm
11. If 20 moles of $\text{PCl}_5(\text{g})$ is added to original equilibrium mixture keeping total pressure constant to 20 atm at same temperature 'T', then ratio of new equilibrium volume to the initial volume V will be -
 (A) 1.5 (B) 2.0 (C) 1.8 (D) 3.0

CHEMISTRY

SECTION-I : (i) (ii) One or more options correct Type

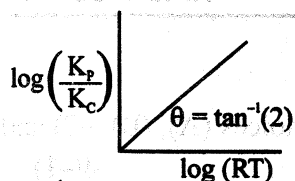
This section contains **9 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. 4(-1)

- How many moles of MnO_4^- will react with 1 mole of ferrous oxalate ion in acidic medium.
(A) $\frac{1}{5}$ (B) $\frac{2}{5}$ (C) $\frac{3}{5}$ (D) $\frac{5}{3}$
- Rate of effusion of two gases 'A' and 'B' under identical condition of pressure and temperature are in ratio of 2 : 1. Calculate ratio of rms velocity of their molecules if T_A and T_B are in ratio of 2 : 1
(A) 2 : 1 (B) $\sqrt{2} : 1$ (C) $2\sqrt{2} : 1$ (D) $1 : \sqrt{2}$
- 10 ml of H_2O_2 solution weighs 10 gm. The solution is diluted to 250 ml. 25ml of of this diluted solution required 40 ml of a M/50 solution of KMnO_4 . Then volume strength of original H_2O_2 solution is-
Given : $5\text{H}_2\text{O}_2 + 2\text{KMnO}_4 + 6\text{HCl} \rightarrow 2\text{KCl} + 2\text{MnCl}_2 + 5\text{O}_2 + 8\text{H}_2\text{O}$
(A) 22.4 V (B) 11.2 V (C) 44.8 V (D) 329.4
- 2.56gm of sulphur $\text{S}_8(\text{s})$ is taken which is in equilibrium with its vapours according to reaction :
 $\text{S}_8(\text{s}) \rightleftharpoons \text{S}(\text{g})$
If vapours occupies 960 ml at 1atm, 300K, then degree of dissociation of $\text{S}_8(\text{s})$ will be -
[R = 0.08 atm L/ mol-K]
(A) 1/4 (B) 1/20 (C) 3/8 (D) 1/2 0.8 x 1/2 = 0.2
- 0.8 M FeSO_4 solution requires 160ml, 0.2M $\text{Al}_2(\text{Cr}_2\text{O}_7)_3$ in acidic medium, Calculate volume of FeSO_4 consumed -
(A) 480 ml (B) 240 ml (C) 720 ml (D) 40 ml
- A x : 1 molar mixture of He and CH_4 is contained in a vessel at 20 bar pressure. Due to a hole in the vessel, the gas mixture leaks out. If the composition of the mixture effusing out initially is 8 : 1. Then calculate the value of x ?
(A) 1 (B) 4 (C) 8 (D) 9
- A solution of $\text{Na}_2\text{S}_2\text{O}_3$ is standardized iodometrically against 0.1262 g of KBrO_3 where BrO_3^- changes to Br^- . This process requires 45 mL of the $\text{Na}_2\text{S}_2\text{O}_3$ solution. What is the molarity of the $\text{Na}_2\text{S}_2\text{O}_3$? [Mw. of $\text{KBrO}_3 = 167$]
(A) 0.2 (B) 0.1 (C) 0.05 (D) 0.01

8. Choose the incorrect statement :

- (A) Rate of effusion of gas is inversely proportional to the molecular weight of gas A-B-D
- (B) Mean free path of gases decreases with increase in temperature at constant volume
- (C) Partial pressure of gas do not changes on addition of another gas in the same container at constant volume and temperature
- (D) Compressibility of real gases are always more than ideal gases

9. Given plot is valid, for which of the following reaction :



- ✓ (A) $2\text{NH}_3(\text{g}) \rightleftharpoons \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$ (B) $\text{NH}_4\text{HS}(\text{s}) \rightleftharpoons \text{NH}_3(\text{g}) + \text{H}_2\text{S}(\text{g})$
(C) $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$ ✓ (D) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s}) \rightleftharpoons \text{CuSO}_4 \cdot 3\text{H}_2\text{O}(\text{s}) + 2\text{H}_2\text{O}(\text{g})$

(iii) Paragraph Type

This section contains **1 paragraphs** each describing theory, experiment, data etc. **2 questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Paragraph for Q.10 to Q.11

During one of its ambitious project "ISRO" has gathered information through its satellites for a planet in another galaxy. As far as structure of atoms is considered, it is very similar to the structure of an atom

in our planet except energy of an orbit in which electron resides is $(E_n) = -\frac{45 \times Z}{(n+1)} \text{ eV/atom.}$

10. Calculate wavelength (in Å) of photon emitted when electron makes transition from 7th excited state to 1st excited state in H-atom of that planet :

- ✓ (A) 1240 Å (B) 12400 Å (C) 310 Å (D) 3100 Å $n=8$

11. If photon of wavelength 7 nm strikes on an atom of hydrogen on that planet, wavelength of ejected electron will be approximately:

- (A) 0.1 Å ✓ (B) 1 Å (C) 5 Å (D) 11 Å

PHYSICAL CHEMISTRY

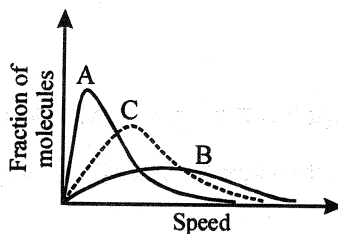
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SECTION-I : (i) Only One option correct Type

This section contains 4 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

3(-1)

- The ratio of wavelength of II line of balmer series and 1st line of Lyman series is-
(A) 4 (B) 2 (C) 3 (D) 16
- When 3 moles of A and 1 mole of B are mixed in 1 litre vessel the following reaction takes place
 $A(g) + B(g) \rightarrow 2C(g)$. 1.5 moles of C are formed. The equilibrium constant for the reaction is -
(1) 0.12 (2) 0.25 (3) 0.50 (4) 4.0
- When an electron makes a transition from $(n + 1)$ to n state, the frequency of emitted radiations (ν) is related to 'n' according to ($n \gg 1$)
(A) $\nu \propto n^{-3}$ (B) $\nu \propto n^2$ (C) $\nu \propto n^3$ (D) $\nu \propto n^{2/3}$
- If two gases have same value of a but different values of b (a and b are vanderwall's constant) which of the following statement is wrong
(A) The gas having smaller value of b has longer compressibility
(B) The gas having smaller value of b will occupy lesser volume
(C) The gas having smaller value of b has lesser compressibility
(D) Both (A) and (B)
- Which among the following equilibrium K_p does not depends upon the initial pressure of reactants
~~(A) $H_2(g) + I_2(s) \rightleftharpoons 2HI(g)$~~ (B) $H_2(g) + Cl_2(s) \rightleftharpoons 2HCl(g)$
(C) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ ~~(D) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$~~
- The curve C is for the gas X at 273 K. Choose the **CORRECT** statement(s) :



- (A) Curve A may be of a heavier gas but at same temperature
(B) Curve B may be of the same gas but at 373K
(C) Curve A may be of the same gas but at 373K
(D) Curve B may be of the lighter gas but at same temperature
- In collection of H-atoms, all electron jump from 5th excited state to ground level finally, without emitting any line in infrared region. Total number of possible different radiations are -
(A) 10 (B) 15 (C) 6 (D) 3

8. **Statement - 1** : The density of an ideal gas doubles when the pressure of the gas is doubled in a closed system.

Statement - 2 : Density of an ideal gas is directly proportional to pressure at constant temperature.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

(iii) Paragraph Type

This section contains **1 paragraphs** each describing theory, experiment, data etc. **2 questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D).

Paragraph for Q.9 to Q.10

According to Le-Chatelier's principle, on applying any external force to disturb equilibrium, then reaction move in that direction in which effect of external force be minimise as far as possible. A container whose volume is V contains an equilibrium mixture that contains **2 moles** each of PCl_5 , PCl_3 and Cl_2 (all gases). The pressure is **3 atm** and temperature is T . A certain amount of $\text{Cl}_2(\text{g})$ is now introduced keeping the pressure and temperature constant until the equilibrium volume changes to $2V$.



9. The moles of Cl_2 that was added is -

(A) 2

(B) 10/7

(C) 8/3

(D) 20/3

10. The equilibrium constant K_p of the reaction

(A) 1

(B) 2

(C) 3

(D) 4

SECTION-IV : (Integer Value Correct Type)

This section contains **1 questions**. The answer to each question is a **single digit Integer**, ranging from **0 to 9** (both inclusive)

4(-1)

1. Wavelength of electron waves in two Bohr orbits is in ratio 3 : 5 the ratio of kinetic energy of electron is 25 : x, hence x is :

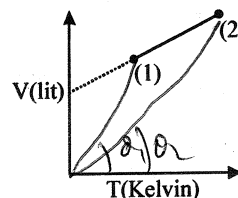
9

SECTION-I : (ii) One or more options correct Type

This section contains **9 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(-1)**

- ✓ 1. For the below graph which of the following combination is correct. ('n' represent mole & 'P' represent pressure)

- I If 'n' is constant throughout then 'P' at 2nd point must be greater than at 1st point.
II If 'n' is doubled in moving from 1st point to 2nd point then 'P' also doubles in the process
III If 'n' is constant then 'P' is maximum at a point (A).



- (A) I is true (B) II are correct
(C) both III & II are true (D) I, II & III all are correct

2. Equal volumes of 0.50M of HCl, 0.25M of NaOH and 0.75M of NaCl are mixed. The molarity of the NaCl solution is.

- (A) 3/4 M (B) 1/3 M (C) 1/2 M (D) 2.00 M

3. A solution is made by dissolving CaBr_2 in water (solvent) such that mass fraction of solute and solvent is same in the solution. The molality of solution is -

- (A) 2.5 m (B) 55.55 m (C) 2 m (D) 5 m

4. Which of the following is correct for 17g/L of H_2O_2 solution-

- (A) Volume strength is 5.6 at 273K and 1 atm
(B) Molarity of solution is 0.5M
(C) 1 ml of this solution gives 2.8 ml O_2 at 273K and 2 atm
(D) The molarity of solution is 2M

5. A straight glass tube as shown, has 2 inlets X & Y at the two ends of 200 cm long tube. HCl gas through inlet X and NH_3 gas through inlet Y are allowed to enter in the tube at the same time and pressure at a point P inside the tube. The distance of point P from X is :

- (A) 118.9 cm (B) 81.1 cm (C) 91.1 cm (D) 108.9 cm

6. Choose the correct statements from among the following -

- (A) A node is a point in space where the wave function (Ψ) has zero amplitude.
(B) Total number of nodes in an orbital is equal to $(n - 1)$
(C) Ψ^2 represents the probability density of finding the electron
(D) Total no. of angular node in $2p_x$ orbital is one.

7. A photosensitive metallic surface has work function $h\nu_0$. If photons of $2h\nu_0$ fall on the surface, the electrons come out with a maximum velocity of 4×10^6 m/sec. If energy of photon is increased to $5h\nu_0$, the maximum velocity of photoelectrons will be.

- (A) 2×10^7 m/sec (B) 8×10^6 m/sec (C) 2×10^6 m/sec (D) 8×10^5 m/sec

8. For the reaction : $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$; $\Delta H = -\text{ve}$. An increase in temperature shows.

- (A) More dissociation of SO_3 and a decrease in Kc
(B) Less dissociation of SO_3 and an increase in Kc
(C) More dissociation of SO_3 and an increase in Kc
(D) Less dissociation of SO_3 and a decrease in Kc

9. The wave function of 3s and 3p_z orbitals are given by

$$\Psi_{3s} = \frac{1}{9\sqrt{3}} \left(\frac{1}{4\pi} \right)^{\frac{1}{2}} \left(\frac{z}{a_0} \right)^{\frac{3}{2}} (6 - 6\sigma + \sigma^2) e^{-\sigma/2}$$

$$\Psi_{3p_z} = \frac{1}{9\sqrt{6}} \left(\frac{3}{4\pi} \right)^{\frac{1}{2}} \left(\frac{z}{a_0} \right)^{\frac{3}{2}} \cdot (4 - \sigma)\sigma e^{-\sigma/2} \cos \theta \quad ; \quad \sigma = \frac{2zr}{na_0}$$

where $a_0 = 1^{\text{st}}$ Bohr radius, $z =$ Charge number of nucleus, $r =$ distance from nucleus

From this we can conclude :

(A) Total number of nodal surface is same for 3s and 3p_x orbitals

(B) The angular nodal surface of 3p_z orbital occur at $\theta = \frac{\pi}{2}$

(C) The radial nodal surface of 3s and 3p_z orbitals are at equal distance from nucleus

(D) 3s electron have greater penetrating power into the nucleus compared to 3p_z electron.

SECTION-II : Matrix-Match Type

This Section contains **1 question**. Question has **four statements** (A, B, C and D) given in **Column I** and five statements (P, Q, R, S and T) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question, against statement B, darken the bubbles corresponding to Q and R in the ORS.

8(0)

Match the Column :

$r_{n,z}$ = Radius of n^{th} orbit of a single electron specie having atomic number 'z'.

$v_{n,z}$ = Velocity of electron in n^{th} orbit of a single electron specie having atomic number 'z'

$E_{n,z}$ = Magnitude of total energy of electron in n^{th} orbit of a single electron specie having atomic number 'z'

$K_{n,z}$ = Magnitude of kinetic energy of electron in n^{th} orbit of a single electron specie having atomic number 'z'

$P_{n,z}$ = Magnitude of potential energy of electron in n^{th} orbit of a single electron specie having atomic number 'z'

$f_{n,z}$ = Frequency of electron in n^{th} orbit of a single electron specie having atomic number 'z'

$T_{n,z}$ = Time period of electron in n^{th} orbit of a single electron specie having atomic number 'z'

Column-I

(A) $\frac{f_{1,2}}{f_{1,1}}$ 4 (B)

(B) $\frac{|E_{3,2}|}{K_{3,1}}$ 4 (B)

(C) $\left(\frac{1}{\sqrt{2}} \right) \left(\sqrt{\frac{P_{2,1}}{E_{3,1}}} \right)$ $\frac{3}{2}$ (R, S)

(D) $\left(\frac{E_{1,4}}{E_{2,2}} \right)$ 16 (T)

Column-II

(P) $\frac{r_{2,3}}{r_{1,3}}$ 4

(Q) $\frac{T_{1,1}}{T_{1,2}}$ 4

(R) $2 \left(\sqrt{\frac{K_{2,3}}{E_{1,2}}} \right)$ $\frac{3}{2}$

(S) $\frac{v_{2,3}}{v_{1,1}}$ $\frac{3}{2}$

(T) $4 \left(\frac{v_{2,5} - v_{2,1}}{v_{2,3} - v_{2,2}} \right)$ 16

SECTION-I : (ii) One or more options correct Type

This section contains **9 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(-1)**

1. What is the correct relation between normality (N) & molarity (M) of $K_2Cr_2O_7$ acting as oxidising agent in acidic medium :

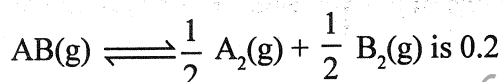
(A) $N = 6 M$ (B) $M = 6 N$ (C) $N = 3 M$ (D) $M = 3 N$

2. For a reaction $A(g) \rightleftharpoons B(g)$ at equilibrium, the partial pressure of B is found to be one fourth of the partial pressure of A. The value of ΔG° of the reaction $A \rightleftharpoons B$ is -

(A) $RT \ln 4$ (B) $-RT \ln 4$ (C) $RT \log 4$ (D) $-RT \log 4$

3. Select correct statements :

- (A) Low pressure is favourable for evaporation of $H_2O(l)$
(B) The degree of dissociation of $CaCO_3(s)$ decreased with increase in pressure
(C) If the equilibrium constant for $A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$ is 25, then equilibrium constant for



$\frac{1}{25}$

$\left(\frac{1}{5}\right)$

- (D) If solid product is added to an equilibrium mixture, the equilibrium will be unaffected.
4. In alkaline medium ClO_2 oxidises H_2O_2 to O_2 and itself gets reduced to Cl^- . How many moles of H_2O_2 are oxidised by 1 mole of ClO_2 ?

(A) 1 (B) 1.5 (C) 2.5 (D) 3.5

5. Which statement about the triple point of a substance is/are **incorrect**

- (A) The triple point for a substance varies with pressure
(B) The three phases (solid, liquid, gas) have same density
(C) The three phases (solid, liquid, gas) are in equilibrium
(D) The three phases (solid, liquid, gas) are indistinguishable in appearance.

6. Iodine molecules dissociates into atoms after absorbing light of wavelength $4500 \text{ \AA}$. If one quantum of radiation is absorbed by each molecule what will be the approx. Kinetic Energy of iodine atom

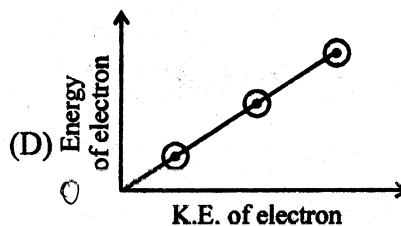
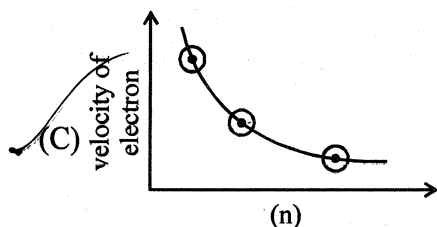
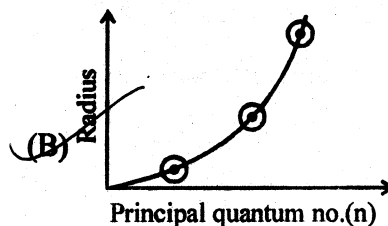
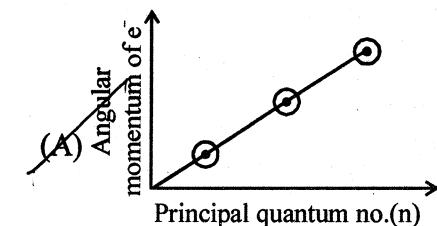
[Given B.E. of $I_2 = 240 \text{ kJ/mol}$, $N_A = 6 \times 10^{23}$]

(A) $8.6 \times 10^{-36} \text{ J}$ (B) $2.17 \times 10^{-20} \text{ J}$ (C) 433 kJ (D) 4.3 J

7. The velocity of electron in the 3rd excited state of a hydrogen atom is [a_0 is Bohr radius]

(A) $\frac{h^2}{3\pi^2 m a_0}$ (B) $\frac{h^2}{3\pi^2 m a_0^2}$ (C) $\frac{8h}{\pi m a_0}$ (D) $\frac{h}{8\pi m a_0}$

8. Select the correct graph for Bohr atom -



9. Which of the following statements is(are) true ?

(A) $\frac{(U_{rms})_{O_2,TK}}{(U_{av})_{O_2,TK}}$, is fixed.

(B) Wavelength of ejected photoelectron can be calculated by $\lambda = \frac{hc}{E}$, where E is kinetic energy of ejected photo electron.

(C) Ozone gas can be absorbed by alkaline pyrogallol solution

(D) Oxygen gas can be absorbed by alkaline pyrogallol solution.

(iii) Paragraph Type

This section contains **1 paragraph** each describing theory, experiment, data etc. **2 questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). 3(0)

Paragraph for Q.10 & Q.11

In three dimension Ψ (wave function) may be expressed in spherical co-ordinate system (r, θ, ϕ)

r = Represents distance of electron from the nucleus

θ = Angle from 'z' axis, varying from 0 to π

ϕ = Angle from 'x' axis 0 to 2π .

Ψ may be represented as

$$\Psi(r, \theta, \phi) = R(r) \cdot A(\theta, \phi)$$

The $R(r)$ is determined by 'n' & 'l'. The $A(\theta, \phi)$ determine how probability change from point to point at a given distance from the centre of the atom.

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10. Which of the following is $R(r)$ part of $3p$ atomic orbital of hydrogen atom [Given $a_0 = 0.529 \text{ \AA}$]

(A) $\frac{2}{(a_0)^{3/2}} e^{-r/a_0}$

(B) $\frac{2}{27} \left(\frac{1}{3a_0} \right)^{3/2} \left(27 - 18 \frac{r}{a_0} + 2 \frac{r^2}{a_0^2} \right) e^{-r/3a_0}$

(C) $\frac{1}{(2a_0)^{3/2}} \left(2 - \frac{r}{a_0} \right) e^{-r/2a_0}$

(D) $\frac{1}{81\sqrt{3}} \left(\frac{2}{a_0} \right)^{3/2} \left(6 - \frac{r}{a_0} \right) \frac{r}{a_0} e^{-r/3a_0}$

11. Angular part of 'H' atom wave equation $A(\theta, \phi) = \frac{1}{\sqrt{4\pi}}$. Hence, atomic orbital is-

(A) ' d_{xz} ' orbital

(B) ' p_x ' orbital

(C) ' p_y ' orbital

(D) s-orbital

(iv) Matching List Type

This Section contains **1 multiple choice questions. Each question has matching lists.** The codes for the lists, have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **4(0)**

12. Match the column :

Column-I

Column-II

(P) 20% (w/w) solution of KOH
(density of solution = 1.02 gm/mL)

(1) 8.64 M

(Q) Solution containing 864 gm
of CaCO_3 in a 1 L.

(2) 3.64 M

(R) Volume of 1.204×10^{24}
molecules of water at 4°C

(3) 5 mL

(S) Volume of 0.2 M NaOH solution
containing 40 mg of NaOH

(4) 36 mL

Codes :

	P	Q	R	S
(A)	4	1	2	3

(B)	2	1	3	4
-----	---	---	---	---

(C)	4	1	3	2
-----	---	---	---	---

(D)	2	1	4	3
-----	---	---	---	---

Handwritten calculations:
 $\frac{40}{40} \times 10^{-3}$
 $10^{-3} = 0.2 \times 2$
 $\frac{40}{40} \times 10^{-3}$

PHYSICAL CHEMISTRY

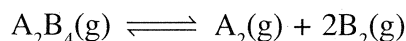
TIME : 30 MIN

SECTION-I : (i) Only One option correct Type

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(-1)**

1. For equilibrium at 300 K, $\text{LiCl}_3 \cdot 3\text{NH}_3(\text{s}) \rightleftharpoons \text{LiCl} \cdot \text{NH}_3(\text{s}) + 2\text{NH}_3(\text{g})$; $K_p = 9 \text{ atm}^2$
In 8.21 L vessel 1.4 mol of $\text{LiCl} \cdot \text{NH}_3(\text{s})$ and 2.8 moles of NH_3 are placed. Moles of $\text{LiCl}_3 \cdot 3\text{NH}_3(\text{s})$ formed will be :
(A) 1.4 (B) 0.7 (C) 0.9 (D) 1

2. A volatile liquid A_2B_4 (M.wt. = 200) has the following equilibrium at an elevated temperature :

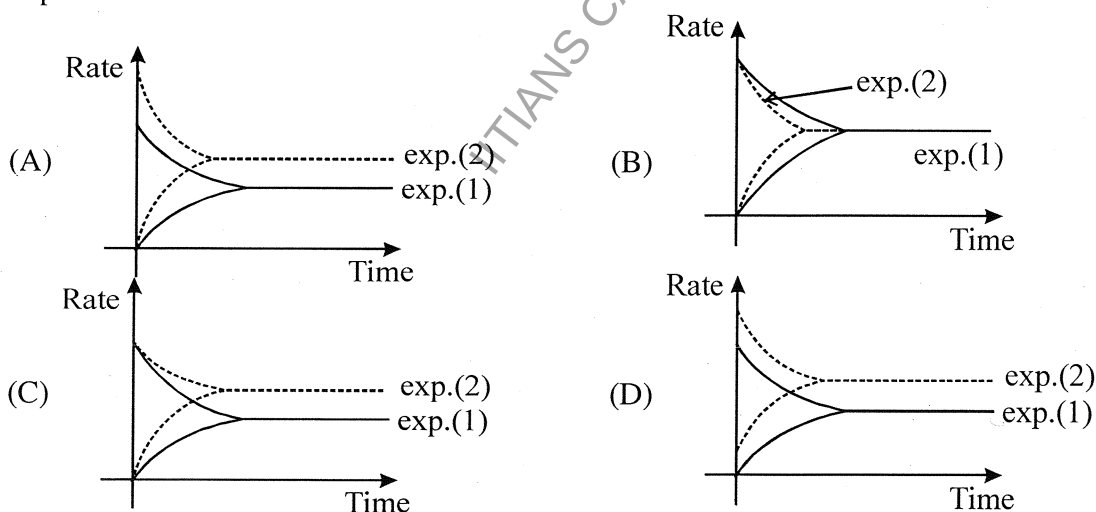


0.01 gm of A_2B_4 , at an elevated temperature displaced 2.27 mL of dry air at S.T.P. The degree of dissociation of A_2B_4 under experimental conditions, is :

- (A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{3}{8}$ (D) $\frac{3}{7}$

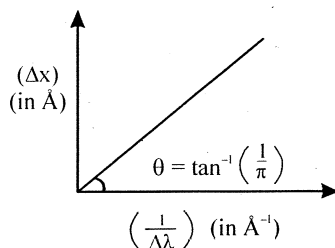
3. In two different experiments equilibrium

$\text{A}(\text{g}) \rightleftharpoons \text{B}(\text{g})$ is established by initially taking reactant only. In first experiment reaction is carried out in absence of catalyst and in second experiment reaction is carried out in presence of catalyst then which of the following curve of rate v/s time for forward and backward reaction hold true in these two experiments -



4. How many milli grams of $\text{Fe}_{0.9}\text{O}$ reacts completely with 10 mL 0.1 M KMnO_4 solution in acidic conditions. (Fe = 56)
(A) 47 (B) 474 (C) 514 (D) 570
5. 4.3 gm of an alkane is burnt in sufficient oxygen. The CO_2 formed reacts completely with 300 ml, 2N NaOH solution producing Na_2CO_3 . The alkane should be
(A) C_3H_8 (B) $\text{C}_{12}\text{H}_{26}$ (C) C_6H_{14} (D) C_2H_6
6. Thallium cyanide, $\text{Tl}(\text{CN})_x$, crystallizes as a simple cubic array of CN^- ions with thallium ions in the cubic holes. What is the oxidation state of thallium in this compound ?
(A) -1 (B) 0 (C) +1 (D) +2

7. An atomic solid crystallizes in bcc lattice if circumference (surface) of atoms at the adjacent corners are separated by 54 pm then edge length of bcc unit cell is ($\sqrt{3} = 1.7$)
(A) 450 pm (B) 108 pm (C) 216 pm (D) 360 pm
8. A graph is plotted between uncertainty in position and inverse of uncertainty in wavelength for an electron. We get a straight line passing through origin. Calculate voltage through which electron is accelerated with -

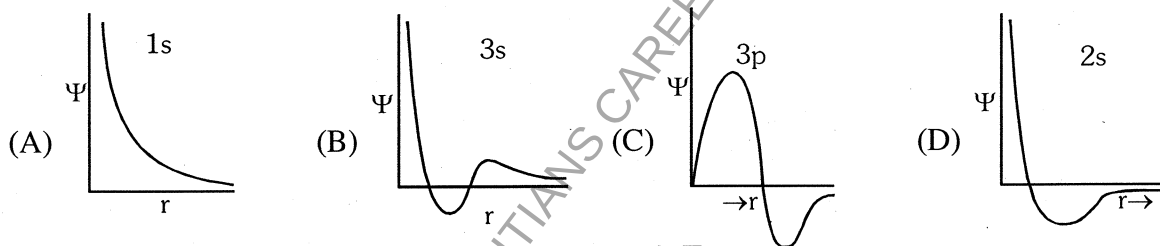


- (A) 150 V (B) 75 V (C) 37.5 V (D) 300 V

(ii) One or more options correct Type

This section contains **1 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. 4(-1)

9. Which of the following graphs of radial part of wave function (Ψ) v/s distance from nucleus is/are correctly labelled ?



(iii) Paragraph Type

This section contains **1 paragraphs** each describing theory, experiment, data etc. **2 questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). 3(0)

Paragraph for Q.10 to Q.11

The vander waals equation was proposed in the year 1873. It was a first step towards taking into account interaction forces which are acting between real gases molecule. The equation of state of the perfect gas refers to a gas consisting of point like items which do not interact with another.

The 'b' correction takes into account the fact that, according to vander waals, the real gas molecules can move not in total volume occupied by the gas, but only in a part of this volume which is defined by subtracting what he called molecules own volume. The correction ' a/v^2 ' - proportional to the square of density takes into account the attraction forces existing between gas molecules.

As far as the critical point, the following conditions are valid :

$$(\partial p / \partial v)_T = 0, (\partial^2 p / \partial v^2)_T = 0$$

The vander Waals equation can be rearranged to a nondimensional form using the reduced parameters.

$$P_r = P/P_c, T_r = T/T_c, V_r = V/V_c$$

In this coordinates the vander waals equation becomes. $\left(P_r + \frac{3}{V_r^2} \right) (3V_r - 1) = 8 T_r$

10. For any vanderwaals gas $P_c = 70 \text{ atm}$ and $V_c = 0.183 \frac{\text{l}}{\text{mol}}$ hence 'a' in $\text{atm-l}^2/\text{mol}^2$ is.....

- (A) 2 (B) 7 (C) 5 (D) 3

11. How many moles of N_2 at 200K and 100 atm will occupy 0.272 litre if $P_r V_r / T_r = 2.21$

- (A) 1 (B) 3 (C) 2 (D) 4

SECTION-IV : (Integer Value Correct Type)

This section contains **3 questions**. The answer to each question is a **single digit Integer**, ranging from **0 to 9** (both inclusive) 4(-1)

1. Three ideal gases H_2 , CH_4 and SO_2 are filled in three rigid containers A, B and C respectively.

$W_{H_2} = 64 \text{ gm}$ $V = 2 \text{ L}$ $T = 200 \text{ K}$	$W_{CH_4} = 64 \text{ gm}$ $V = 1 \text{ L}$ $T = 400 \text{ K}$	$W_{SO_2} = 128 \text{ gm}$ $V = 2 \text{ L}$ $T = 1600 \text{ K}$
A	B	C

the ratio of most probable speeds of molecules in all containers is $x : y : z$ then calculate the value

of $\left(\frac{x+y}{z} \right)$.

2. The minimum number of waves made by a Bohr electron in an orbit of maximum magnetic quantum number 3 will be -
3. Calculate the density (in gm/cm^3) of NaCl type ionic solid ($M_w = 75 \text{ gm/mol}$) if distance between two nearest cations is $250\sqrt{2} \text{ pm}$ (avogadro number = 6×10^{23})